

Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT Sparse phasor measurement units leave two different questions unresolved: whether the unmeasured voltage field can be reconstructed and whether the physical sources of an event can be distinguished. We separate those questions with a hybrid differential-algebraic model. Functional observability governs the requested voltage target; finite-horizon separation between candidate response manifolds governs event cardinality and source support. We test this distinction on the IEEE 39-bus system with eight voltage/current PMUs. A causal model-based estimator reconstructs the 31 unmeasured bus voltages with 0.009634% mean relative error under matched operation. A separate physical likelihood evaluates 137 load-intervention hypotheses, comprising no event, every single source, and every two-source support. On 3488 fresh nonlinear trajectories, exact-support Top-1 ranges from 24.4% for weak pairs to 100.0% for finite pairs. The weak cases reveal the central limitation: posterior evidence can identify that two sources are active before it can identify which two. Profiled response distance explains this ambiguity better than support-only geometry, while longer observation increases separation without removing severe signature collinearity. The results support matched-model state reconstruction and simultaneous load-source inference. They do not establish a multi-family joint estimator, changing-PMU operation, unknown-onset inference, or field transfer.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet transmission networks instrument only a fraction of their buses and branches [1], [2]. An event may start outside the measured region and still alter every available PMU. Missing frames and corrupted channels create a second problem because they can resemble a physical transient. A useful diagnosis must therefore answer separate questions: what is the hidden network state, what changed physically, and which measurements can still be trusted?

Dynamic estimators combine network models with synchrophasors [2], [3]; sparse-PMU classifiers identify events and faulted assets [4], [5]; and recent learning methods reconstruct hidden trajectories before localization [6], [7]. These capabilities do not imply one another. In particular, a low state-reconstruction error says nothing about whether two

candidate sources produce distinguishable measurements.

This distinction changes how the problem must be evaluated. A generation change may be visible throughout the network while its source remains ambiguous. A dropped PMU identifies the failed measurement device, yet the lost channel can make a simultaneous physical event harder to localize. Treating both cases as entries in one multiclass label hides the relevant failure mechanism and mixes incompatible denominators.

Our central idea is to separate target recovery from source separation inside the same physical model. A candidate event specifies a family, a sparse support, an onset, and signed severity. Measurement availability and corruption remain separate variables. Each candidate generates a counterfactual DAE response. After admissible state and integrity directions are profiled, the distance between candidate response manifolds measures the information available for source dis-

crimination [8], [9]. Zero distance means that two candidates intersect; a small positive distance means that they are distinct but difficult to separate with the available horizon and noise.

We test that argument in two stages. The state study asks what eight voltage/current PMUs recover, how the estimate fails when its operating point is stale, and whether a constrained physical recentring repairs that failure. The event study then evaluates all zero-, one-, and two-source load supports without fitting a bus-label classifier. A retrospective geometry audit and a longer-horizon diagnostic explain the weak-source failures. Finally, an earlier supervised ANDES model provides a contrasting test: it performs well when a source appears in training but collapses under complete source holdout. These studies use separate targets and denominators.

The contributions are:

- 1) a hybrid DAE formulation that keeps physical interventions and measurement integrity compositional, and represents simultaneous sources by a sparse support rather than a new class label;
- 2) a finite-horizon analysis that separates functional state recovery, sparse event injectivity, and the conditioning supplied by additional PMUs or observation time, and connects profiled separation to Gaussian error and support-recovery limits; and
- 3) a prospective 137-hypothesis load-source test that connects exact-support recovery to response-manifold geometry, together with independent state-reconstruction and source-holdout evidence.

The evidence yields one consistent picture. Sparse PMUs can recover the requested voltage field under a matched model, but operating-point error must be corrected physically. For simultaneous load events, source severity controls whether the posterior resolves only cardinality or the exact support. Longer observation adds information, yet nearly collinear signatures remain difficult. The evaluated method therefore supports matched load interventions with at most two sources. It does not yet solve the full multi-family problem shown in Fig. 1.

II. PRIOR WORK AND RESEARCH GAP

A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with synchronized measurements; model uncertainty and measurement availability remain persistent constraints [2]. Robust filters address non-Gaussian measurement errors [10], [11], whereas joint input/state estimators recover exogenous inputs in linear systems [12], nonlinear DAEs [3], [13], and event-triggered or distributed power networks [14], [15]. These estimators do not define a multi-family physical-source bank with a simultaneous integrity state.

Learning methods reconstruct states from sparse measurements by encoding graph structure or power-flow relations.

Examples combine graph learning with physics-based estimates [16], outlier-resistant estimation and outage identification [17], or topology-aware reconstruction under PMU failures and bad data [18]. Robust reconstruction does not, by itself, identify the physical source.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification uses subspaces, engineered features, transfer learning, and graph models [19]–[22]. Label-light field-stream detection and scarce-label transfer target event presence or family [23], [24]; physical-source recovery is distinct. Prior localization spans joint classification/localization [4], sparse-PMU fault, generation-loss, and outage methods [5], [25]–[27], group-sparse physical dictionaries [28], and topology-constrained graph losses [29].

Model-based outage inference provides the closest classical comparison. Phasor angles support line-outage detection [30]; uncertain pre-event state can be marginalized jointly with outage identity and sensor placement [31]; and sequential likelihoods handle stochastic operating variation [32], [33]. Matched-subspace detectors profile linear nuisance through a generalized likelihood ratio [9]. These are the likelihood and nuisance-elimination precedents used here. Our load contract instead uses a sparse voltage/current response dictionary and known onset; it is not a quickest-detection comparison.

Open-set recognition differs from source holdout. Siamese boundaries and generative evidence reject events outside learned classes [34]–[36]. Source generalization instead ranks a physically enumerable but unlabeled asset as a counterfactual source.

The pmuBAGE collection supports standardized analytics but does not certify source holdout [37]. Integrity is a separate variable: robust recovery distinguishes excursions from outliers [38], while utility examples include missing, locked, quantized, and timing-affected synchrophasors [39].

The unresolved issue is the evaluation contract. New replicas from trained assets measure interpolation, not source generalization; nuisance variation can also alter conclusions [40]. Reconstruction followed by localization is prior art [6], [7], but not under the combined held-source, diagnosability, integrity, and mismatch tests used here.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [41], [42]; nonlinear extensions require the observation fibers to determine the requested functional locally, not merely a favorable Jacobian at one point [43]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [44]; fault detectability asks whether a fault can be separated while disturbances are decoupled [45]. Sparse strong observability extends the question to corrupted sensor sets [46], [47], and attack identification characterizes when adversarial input supports are distinguishable from sys-

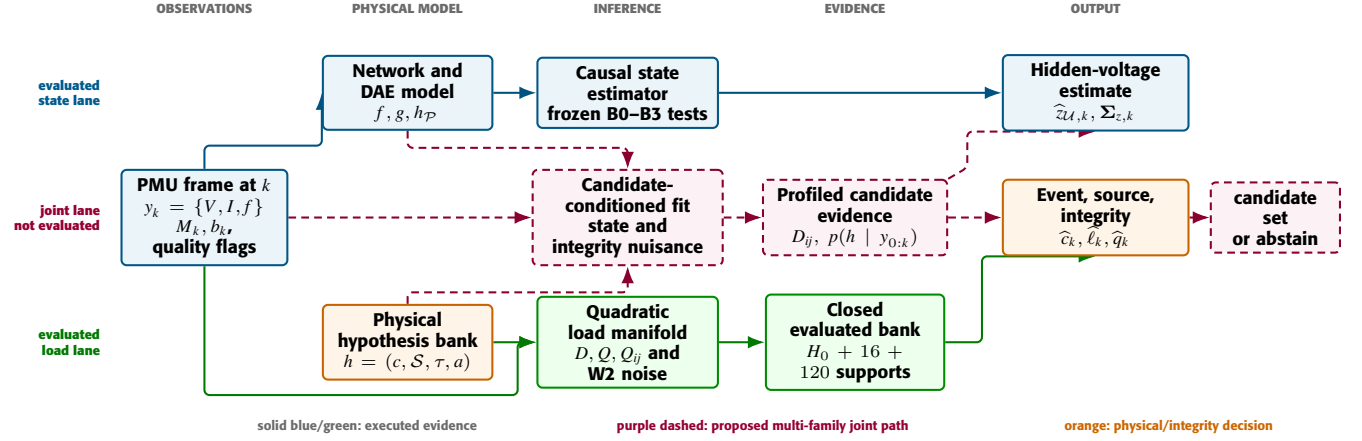


FIGURE 1. Scientific argument and validation boundary. The evaluated state path asks what the sparse PMUs recover. The evaluated load-event path asks which candidate supports the measurements distinguish. The dashed middle path is the intended multi-family integration; it remains outside the present evidence.

tem behavior [48]. Eriksson *et al.* quantify pairwise diagnosability for stochastic linear descriptor models [8]. Likelihood inference with singular first-order information also has established nonregular asymptotics [49]; therefore, a vanishing projected tangent is not by itself a universal impossibility result. PMU placement based on empirical observability Gramians treats sensor value as task dependent [50], [51]. For event sources, Jamei *et al.* derive placement-dependent fault-localization limits [52]; Conte *et al.* partition a radial grid into indistinguishable line clusters [53]; and Yildiz and Abur optimize sparse placement for event detection and identification [54]. These results distinguish hidden-state reconstruction, input reconstruction, and pairwise source separation. They also rule out a general optimal-placement claim from one prescribed eight-PMU set.

D. REMAINING TECHNICAL GAP

Table 1 shows why one accuracy value cannot close the gap. The state path must specify the requested function, and the event path must expose indistinguishable sources before any classifier is trusted. A source-generalization test must also remove the physical asset from every source-labeled fitting step. Under that contract, learning is useful only if it corrects model discrepancy or reduces candidate-search cost without memorizing asset identity.

The experiments cover only part of this intersection. PowerDynamics supports hidden-voltage estimation, physical re-centering, and prospective load-source inference. ANDES documents what a supervised representation loses under source holdout. A common multi-family experiment with state and integrity nuisance remains future work.

III. STATE, EVENT, AND SOURCE PROBLEM

A. ONE PHYSICAL MODEL, TWO INFERENCE QUESTIONS

Let $x(t)$ and $z(t)$ denote differential and algebraic states, $q(t)$ the network mode, and θ the operating parameters. A physical

candidate is

$$h = (c, \mathcal{S}, \tau, a, d), \quad (1)$$

where c is the event family, $\mathcal{S}$ its bus or branch support, τ the onset, a the signed severity, and d the temporal profile. The hybrid plant and measurement boundary are

$$\dot{x} = f_h(x, z, u; \theta) + w_x, \quad \mathbf{0} = g_h(x, z, u; \theta), \quad (2)$$

$$y_k = M_k[h_{\mathcal{P}_k}(x_k, z_k, q_k; \theta) + b_k] + v_k. \quad (3)$$

The event enters the physical equations through h . In contrast, M_k records channel availability and b_k records measurement corruption. A physical event and a failed PMU can therefore coexist without becoming a new event class.

For a scheduled impulse-free intervention, the physical switch preserves the differential state and re-equilibrates the algebraic state in the post-event mode:

$$x(\tau^+) = x(\tau^-), \quad g_{h^+}(x(\tau^-), z(\tau^+), u; \theta^+) = \mathbf{0}. \quad (4)$$

This right limit is not necessarily the array stored inside a simulator callback. The frozen trajectories begin at the first retained 30-Hz frame, so they validate the sampled response rather than the callback microstate. Supplementary Appendix D proves the local consistency expansion and separates the direct algebraic PMU signature from its subsequent dynamic propagation.

The network map uses $I = Y_{\text{bus}}V$ and $S_i = V_i I_i^*$. A fault changes a shunt, a line outage changes Y_{bus} , and generation or load events change the corresponding injections. PMU currents use the instrumented branch terminal and orientation. Supplementary Appendix B gives the AC power equations, transformer and terminal-current conventions, index-one reduction, and frequency/ROCOF definitions. Those details determine the likelihood, but they do not change the two inference questions posed here.

The first question asks for a function of state. With PMUs at

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad (5)$$

TABLE 1. Research gap by methodological family. The full paper-by-paper capability matrix is reported in Supplementary Appendix A.

Family	Established capability	Remaining issue for this problem	Role here
Dynamic state and unknown-input estimation [3], [12], [13]	Reconstructs states or unknown inputs from model and PMU histories	Does not usually define a closed physical event-source bank with compositional integrity	Defines the state-reconstruction path and its nuisance variables
Learned event localization [4], [5], [29]	Detects and localizes represented event families under sparse sensing	Source-label training can confound asset memorization with source generalization	Supplies the supervised baseline and held-source stress test
Model-based event inference [30]–[32]	Ranks physical outage hypotheses and can marginalize operating uncertainty	Published contracts are event-specific and do not establish the present simultaneous-load problem	Motivates physical candidate likelihoods and nuisance profiling
Observability and placement [43], [52], [54]	Characterizes state recovery or event separation for a declared sensing task	Rank alone does not quantify finite-horizon support ambiguity	Supplies the target-recovery and diagnosability tests
This work	Reconstructs a requested hidden-voltage field and ranks all load supports with $K \leq 2$	The multi-family state/event/integrity posterior remains untested	Connects separate evaluated paths through one information-based formulation

the requested target is the complex voltage at the remaining 31 buses. The state estimator uses rectangular voltage and controlled terminal-current channels. The second question asks which physical candidate generated the observed response. For the evaluated load family,

$$\mathcal{S} \subseteq \mathcal{L}, \quad |\mathcal{L}| = 16, \quad K = |\mathcal{S}| \leq 2, \quad (6)$$

which yields one no-event hypothesis, 16 single-source hypotheses, and 120 two-source supports.

B. PHYSICAL EVENTS AND MEASUREMENT INTEGRITY

The historical event bank has eight abnormal labels, but the label number is not the scientific variable. Table 2 rewrites the bank as four physical interventions, two integrity states, and two prescribed compositions. This representation matters because a missing packet identifies an integrity source directly while reducing the information available for physical-source inference.

The desired decision consists of the hidden-voltage estimate, its uncertainty, the physical family and support, and the integrity state. These components use different correctness tests. Hidden-voltage relative error is averaged over a complete trajectory and the 31 requested buses. Event detection, family classification, exact-support recovery, and integrity localization retain their own populations. A joint decision is correct only when every applicable component is correct.

Causality and generalization are also explicit. A causal estimate at time k may use only $y_{0:k}$; fixed-lag or complete-window results are retrospective. A held-source test removes the physical asset from every source-labeled fitting and tuning step. Confidence intervals use trajectories, event parents, or rebuilt plants as independent units, never individual 30-Hz samples. Supplementary Appendix C records the exact metrics and event operators.

IV. WHAT SPARSE MEASUREMENTS CAN DETERMINE

The analysis separates two questions before an estimator is chosen. The first asks whether the measured history determines the requested hidden-voltage function. The second asks whether it distinguishes competing event supports. Both depend on the sensing geometry, but they fail for different reasons.

A. RECOVERING A REQUESTED VOLTAGE FUNCTION

Linearizing the index-reduced DAE around a verified equilibrium gives

$$\begin{aligned} \delta x_{k+1} &= A_d \delta x_k + w_k, & \delta y_k &= C \delta x_k + v_k, \\ \delta z_{\mathcal{U},k} &= L_{\mathcal{U}} \delta x_k. \end{aligned} \quad (7)$$

Here $\delta x_k \in \mathbb{R}^{114}$, the 32 measured channels form δy_k , and the 62 rectangular voltage components at the 31 unmeasured buses form the target. For a horizon of N transitions, let

$$\mathcal{O}_N = \text{col}(C, CA_d, \dots, CA_d^N), \quad L_r = L_{\mathcal{U}} A_d^r. \quad (8)$$

Proposition 1 (Finite-horizon functional reconstruction): The noise-free measurement history through N uniquely determines the requested voltage at time $r \leq N$ if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \quad (9)$$

The condition is weaker than full-state observability: unobservable state directions are harmless when they do not change the requested voltage. Under Gaussian noise, the same row-space condition yields the minimum-variance linear unbiased target estimate; its covariance depends on singular values and target alignment, not rank alone [41], [42]. Supplementary Appendix D gives the proof, the covariance expression, and the nonlinear constant-rank qualification [43].

The numerical audit illustrates why that qualification matters. The original implementation formed $\mathcal{O}_N^T \mathcal{O}_N$ and applied a relative eigenvalue threshold. Its pseudo-rank rises and

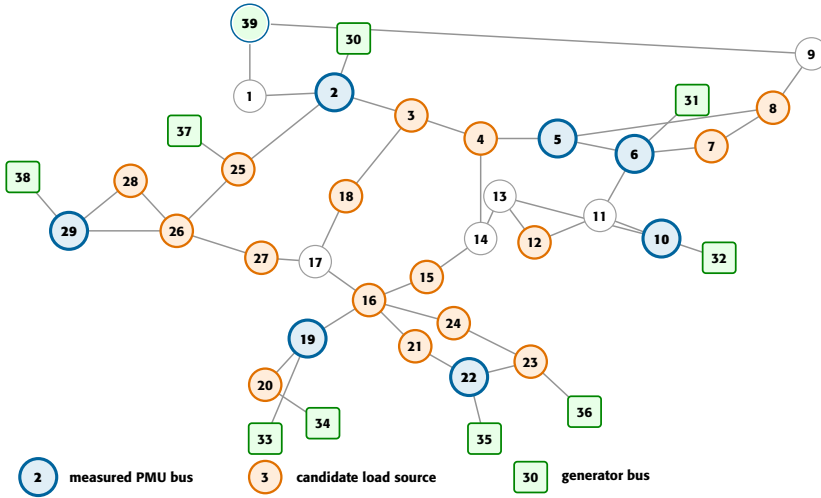


FIGURE 2. Sensing and candidate geometry on the IEEE 39-bus system. Blue circles mark the eight PMU buses, orange circles the 16 evaluated load-source buses, and green boxes generator buses. The drawing preserves the audited 46-branch connectivity but does not encode geographic or electrical distance.

TABLE 2. Event ontology and historical ANDES inventory. Each source configuration has five replicas. Physical and integrity sources remain separate outputs.

ID	Operational meaning	Physical intervention	Physical source	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
Total					690

then falls as samples are added, which an exact observability rank cannot do. Figure 3 therefore separates two conclusions. Longer history improves numerical conditioning, but the tested computation does not certify full-state or exact functional observability. The nonlinear estimator is evaluated independently.

The bus-level residual from this audit was frozen before the nonlinear TEST trajectories were scored. Its association with empirical bus error is used only as an exploratory theory-to-data check because the projector remains threshold sensitive. An SVD-based, noise-tied rank audit is still required for a structural certificate.

B. DISTINGUISHING EVENT SUPPORTS

Source inference introduces a different nuisance space. Under candidate h , stack the available measurements over H samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H}\xi_h + \mathcal{E}_{S,H}\beta_h + \varepsilon_H, \quad (10)$$

where $s_{h,H}$ is the physical response, ξ_h an admissible initial-state deviation, and $\mathcal{E}_{S,H}\beta_h$ corruption on integrity support S .

Each candidate therefore defines an affine response manifold. For candidates i and j , their profiled separation is

$$D_{ij,S}(H) = \frac{1}{2} \min_{\xi_i, \xi_j, \beta} \|s_{i,H} + \mathcal{O}_{i,H}\xi_i - s_{j,H} - \mathcal{O}_{j,H}\xi_j + \mathcal{E}_{S,H}\beta\|_{R_H^{-1}}^2. \quad (11)$$

This distance profiles the nuisance directions that either candidate can use to explain the same measurements. It specializes descriptor-model diagnosability and matched-subspace testing to the PMU event bank [8], [9].

Proposition 2 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the two candidate manifolds intersect. At an intersection they induce the same Gaussian measurement law, so no rule can distinguish them uniformly over the declared nuisance set.

A positive distance establishes local separation, not an easy decision. Small distance or near-collinear signatures amplify noise and delay support resolution. Supplementary Appendix D proves the proposition, treats unknown sparse corruption, and shows that adding a conditionally independent PMU cannot reduce the profiled margin when the nuisance contract is fixed.

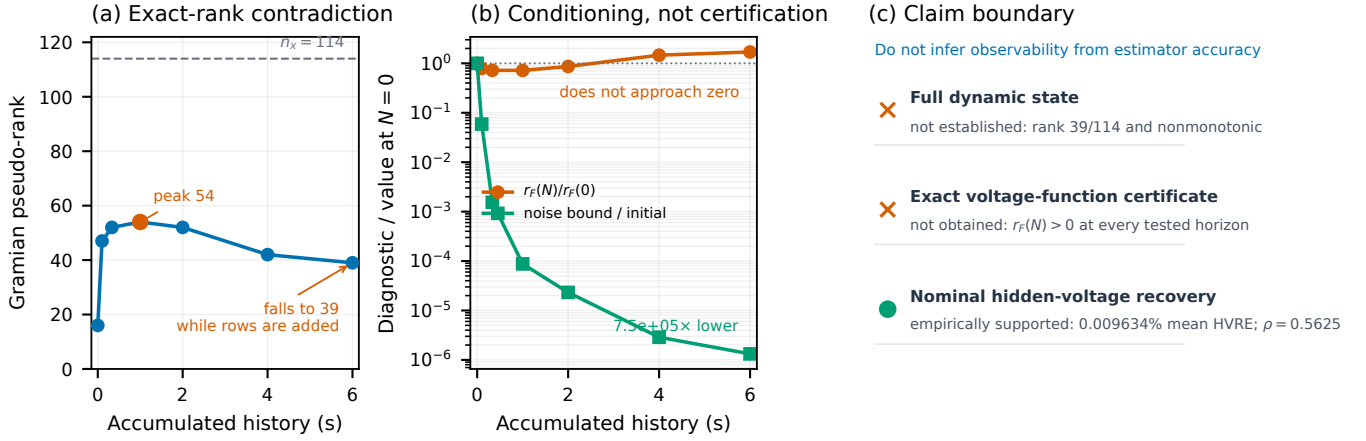


FIGURE 3. Structural audit and claim boundary. Adding history lowers the target-noise bound, but the relative-threshold pseudo-rank is nonmonotone and the target residual remains nonzero. The audit therefore motivates a conditioning analysis; it does not prove full-state or exact hidden-voltage observability.

The infinitesimal version makes the first-order boundary explicit. Let g_* be the event-response tangent, J_c the tangent of the closest competing nuisance manifold, and $W^T W = R_H^{-1}$. With

$$\gamma_1^2 = \|(I - P_c) W g_*\|_2^2, \quad P_c = (W J_c)(W J_c)^\dagger, \quad (12)$$

the corresponding local version of (11) satisfies $D_{*c}(a) = a^2 \gamma_1^2 / 2 + o(a^2)$. Thus $\gamma_1 > 0$ gives first-order separation after nuisance fitting; $\gamma_1 = 0$ only makes the first-order test inconclusive. Relative curvature can then produce an a^4 squared-distance law. The full statement and proof are in Supplementary Appendix D. The present experiments do not yet test the convergence of $2D_{*c}(a)/a^2$ to γ_1^2 .

For multiple events, let G collect the first-order event signatures, N the nuisance tangent, let P_A denote the orthogonal projector onto $\text{range}(A)$, and define $\bar{G} = (I - P_{WN})WG$ the event dictionary after nuisance removal. All first-order event information that survives worst-case nuisance fitting lies in $\mathcal{D} = \text{range}(\bar{G})$. Hence $d_{\mathcal{D}} = \text{rank}(\bar{G})$ is the intrinsic linear diagnostic dimension, not the number of retained PMU channels or reconstructed states. Supplementary Appendix D proves that coordinates on $\mathcal{D}$ preserve every profiled Gaussian likelihood ratio when candidates share the same nuisance range. For true support S , severity direction α_S , and competing support C , the support-to-support margin is

$$\gamma_{S \rightarrow C}^2(\alpha_S) = \|(I - P_{\bar{G}_C}) \bar{G}_S \alpha_S\|_2^2. \quad (13)$$

Equation (13) is exact for signed amplitudes or an interior parameter point. At a physical boundary, let $T_{\mathcal{A}_c}(\mathbf{0})$ be the tangent cone of the competitor's admissible amplitude domain. The first-order physical margin is instead

$$\gamma_{\text{phys}}^2 = \min_{u, \beta_C \in T_{\mathcal{A}_c}(\mathbf{0})} \|WG_S \alpha_S - WN u - WG_C \beta_C\|_2^2. \quad (14)$$

Because the cone is contained in its span, $\gamma_{\text{phys}} \geq \gamma_{S \rightarrow C}$; zero relaxed margin does not prove ambiguity for a one-sided event. Shared sources cancel exactly only when the matched amplitude is feasible and the relevant constraint is

inactive. Competitor enrichment still cannot increase the margin, and a nested superset still has zero margin when its added amplitude admits zero. Geometry can therefore rank non-nested explanations, but cardinality among nested supports requires a model-selection penalty or integrated evidence. Supplementary Appendix D proves the subspace and cone statements.

For persistent load events, subtract the common no-event trajectory and write the T -sample response as

$$Y_T = \mathcal{H}_T a + \varepsilon_T. \quad (15)$$

Let $\mathcal{A}_{2K}$ contain vectors with at most $2K$ nonzero entries.

Proposition 3 (Finite-horizon sparse event injectivity): Every event with at most K active sources is uniquely determined in the noise-free model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{A}_{2K} = \{\mathbf{0}\}. \quad (16)$$

Stable amplitude estimation also requires a well-conditioned support Gramian.

The proposition distinguishes uniqueness from finite-information difficulty. With prefix-consistent causal whitening, adding samples cannot reduce the profiled separation between two fixed supports. Separation grows when their steady signatures differ and can saturate when only a finite transient distinguishes them. The experiments therefore ask two questions in order: does the posterior identify how many sources are active, and does the remaining separation identify which sources? Full proofs and the temporal cases appear in Supplementary Appendix D.

The separation also has an operational meaning under the corresponding simple equal-covariance Gaussian quotient experiment. If d is the nuisance-profiled Mahalanobis distance, then

$$P_e^* = \Phi(-d/2), \quad D_{\text{KL}} = d^2/2. \quad (17)$$

Thus a local order- k pair satisfying $d_T^2(a) = T\mathcal{I}_k|a|^{2k} + o(T|a|^{2k})$, with $\mathcal{I}_k > 0$, has the resolution scale

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}} \right]^{1/k}. \quad (18)$$

First-order and curvature-only separation therefore improve as $T^{-1/2}$ and $T^{-1/4}$, respectively. Equation (17) is exact for the stated quotient experiment, not for an amplitude-marginalized nonlinear support mixture.

Two limits prevent accuracy from being interpreted without a severity and candidate-set contract. If an added active amplitude may approach zero, its law approaches the nested lower-cardinality law, so uniform exact-support recovery is impossible without a minimum-severity condition or abstention. For M equiprobable supports, Fano's inequality further gives

$$P_e \geq \left[1 - \frac{\bar{D}_{\text{KL}} + \log 2}{\log M} \right]_+, \quad (19)$$

$$\bar{D}_{\text{KL}} = \frac{1}{M^2} \sum_{i,j} D_{\text{KL}}(P_i \| P_j).$$

which separates algorithmic failure from insufficient physical information [55], [56]. Supplementary Appendix D states the assumptions, proofs, and the resulting necessary horizon bound.

V. TWO EVALUATED INFERENCE PATHS

Figure 1 separates the implemented paths from the intended joint model. The blue path reconstructs a requested voltage function under one physical mode. The green path compares load-event supports through simulated physical responses. They share the DAE and PMU operator, but they do not share a fitted posterior in the present experiments.

A. CAUSAL STATE RECONSTRUCTION

The state experiment changes one information source at a time. B0 returns the equilibrium voltage. B1 conditions that equilibrium on the current PMU frame. B2 propagates the state causally and then applies the PMU correction. B3 is a retrospective fixed-lag smoother used only to test whether later measurements add useful information. This sequence isolates where each improvement comes from.

For the local model in (7), B2 uses the standard Kalman recursion [57]:

$$\begin{aligned} \hat{x}_{k|k-1} &= A_d \hat{x}_{k-1|k-1}, \\ P_{k|k-1} &= A_d P_{k-1|k-1} A_d^T + Q, \\ K_k &= P_{k|k-1} C^T (C P_{k|k-1} C^T + R)^{-1}, \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k \nu_k. \end{aligned} \quad (20)$$

The hidden-voltage estimate is $z_{U,0} + L_U \hat{x}_{k|k}$. Only measurements through k enter B2. B3 follows the Gaussian backward recursion [58] and is explicitly retrospective.

Two checks separate point accuracy from uncertainty. First, a temperature map fitted on independent calibration trajectories rescales the two rectangular voltage components without

changing the state estimate. Second, an observed-innovation statistic tests whether the nominal model has become inadequate. Neither operation repairs the physical model. Their complete equations and negative discrepancy ablations appear in Supplementary Appendix E.

When the operating point is stale, the physical-recentering estimator rebuilds a feasible equilibrium instead of shifting measured voltages by an unconstrained offset. It solves

$$(\hat{V}, \hat{d}) = \arg \min_{V, d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{\text{AC}}(V, d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2, \quad (21)$$

where e_y uses only measured voltage and terminal-current phasors, g_{AC} enforces AC balance and bus conventions, and d collects admissible load and generation changes. Development data fix the weights and continuation rule before TEST. Hidden voltages and the true nuisance vector never enter the solve.

B. PHYSICAL LIKELIHOOD FOR LOAD-EVENT SUPPORTS

The event path begins with a physical question: what would the measured PMUs have recorded if a candidate load bus had changed? For each of the 16 buses, centered nonlinear simulations form the unit response

$$D_g = \frac{y_g(+\epsilon) - y_g(-\epsilon)}{2\epsilon}, \quad \epsilon = 0.005. \quad (22)$$

Symmetric simulations also estimate source self-curvature Q_i and pairwise interaction Q_{ij} . The mean response for support S is

$$\mu_S(a) = \sum_{i \in S} (a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j \in S \\ i < j}} a_i a_j Q_{ij}. \quad (23)$$

The mixed term matters physically: two simultaneous interventions need not equal the sum of two isolated responses.

After subtracting the normal mean, the residual follows $r = \mu_S(a) + \varepsilon$ under the frozen Gaussian covariance. Each support receives normalized evidence by integrating over its active amplitudes,

$$Z_S = \int \mathcal{N}(r; \mu_S(a), \Sigma) \prod_{i \in S} p(a_i) da_S, \quad (24)$$

$$p(S | r) \propto Z_S \frac{p(|S|)}{\binom{16}{|S|}}.$$

Gauss-Hermite quadrature evaluates every support; independent Gauss-Kronrod calculations check representative two-dimensional integrals. Priors, covariance estimation, quadrature order, and calibration checks are documented in Supplementary Appendix E.

The amplitude integral is structurally necessary, not decorative. A superset can reproduce any smaller support by driving its additional amplitudes to zero, so its pure profiled geometric distance can also be zero. Integrated evidence accounts for the volume of amplitude values that fit the data, while $p(|S|)$ and the within-cardinality prior state the sparsity

contract. Cardinality is therefore a Bayesian model-selection question; it cannot be recovered from best-fit distance alone.

The evaluated load bank uses a proper signed Gaussian amplitude prior because load changes may increase or decrease. A one-sided fault, trip, or outage instead requires a proper truncated slab and boundary-aware evidence. In the local single-amplitude Gaussian model, Supplementary Appendix E shows that a half-normal slab multiplies the signed Bayes factor by an exact directional factor $2\Phi(z)$. This extension is part of the common estimator design; it is not an evaluated multi-family result.

The posterior is read in a fixed order. First, $p(K | r)$ asks whether no source, one source, or two sources are active. Conditional support ranks then ask which buses are involved. Amplitude intervals answer how large the changes are. This order is essential under limited observability because cardinality can be well supported while several source pairs remain plausible.

The prospective experiment evaluates all 137 supports at a known onset and fixed PMU set. It conditions on the nominal initial state and assumes no corrupted channels. Thus it evaluates the sparse load specialization of Proposition 3, not the full state-and-integrity nuisance model in (10).

C. BOUNDARY OF THE JOINT FORMULATION

The statistical object in the complete formulation is the candidate-conditioned joint posterior and normalized evidence, with state and integrity nuisance modeled under each physical hypothesis. Iterated smoothing, Gauss–Newton steps, moving-horizon optimization, or generalized EM would be numerical realizations of that object, not different scientific estimators. Changing the solver cannot change the declared priors, candidate set, causal data boundary, or evidence normalization.

A learned component could then correct candidate residuals or reduce the search, but it must not consume source identity as a shortcut. It must also preserve the directions that distinguish critical physical candidates. After whitening and nuisance removal, a rank- q discrepancy basis Q_D with variance τ^2 retains the worst-case fraction

$$\rho = 1 - \frac{\tau^2}{1 + \tau^2} \sigma_{\max}^2(Q_U^T Q_D) \quad (25)$$

of the information in a protected discriminant subspace Q_U . Supplementary Appendix D derives the corresponding finite-variance rule using the full Gaussian likelihood; maximizing residual accommodation alone would omit its log-determinant penalty. The required common-contract ablation remains learned-only, physics-only, physics plus learned discrepancy, and diagnosability-aware inference with abstention. None of those integrations is presented as an executed result here; Supplementary Appendix E records the full generative objective so the proposed extension is unambiguous.

VI. EXPERIMENTAL QUESTIONS AND FROZEN EVIDENCE

The experiments follow the logic of the paper rather than the order in which the software was developed. Each re-

search question has its own independent unit, frozen selection boundary, and claim limit (Table 3). The complete manifests, seeds, acceptance gates, simulator versions, and secondary campaigns appear in Supplementary Appendix F.

A. RQ1 AND RQ2 STATE-RECONSTRUCTION CONTRACTS

The primary plant is the package-defined IEEE 39-bus PowerDynamics benchmark [59]. Before scoring estimator error, we verified the power-flow translation, terminal-current orientation, equilibrium, local modes, nonlinear trajectories, and first-order response. These checks establish numerical consistency with the implemented benchmark; they do not make the tutorial model an authoritative external grid.

RQ1 compares B0–B3 on the same 3-s trajectories. Model parameters and covariance choices are fixed before TEST, and the 31 requested buses exclude all eight observed buses. Separate calibration trajectories fit uncertainty corrections. RQ2 then rebuilds the plant under seven physical mismatch families while keeping the nominal estimator frozen. A distinct held-out operating-point campaign tests the constrained recentering in (21). Hidden voltages are revealed only after each solve, and rejected solves remain in the planned denominator.

B. RQ3 SIMULTANEOUS LOAD-SOURCE CONTRACT

Development data select the normal covariance and likelihood before the prospective support test. The final bank freezes D_i , Q_i , Q_{ij} , the priors, onset, and quadrature, then generates new nonlinear trajectories for H_0 , every single source, and every source pair. Event identifiers do not overlap the historical development manifests. Repeated noise realizations produce separate decisions but are not counted as new physical-system trajectories.

The primary outcome is exact support, evaluated only after cardinality. A retrospective weak-regime audit asks which geometric distance explains errors without refitting the likelihood. A separate 120-frame study checks physical-manifold fidelity, information accumulation, and runtime. Because that study uses only one operating point and 32 validation paths, it is a diagnostic rather than a prospective recovery result.

C. RQ4 HISTORICAL BASELINE AND EVENT SEMANTICS

The ANDES campaign [60] preserves the earlier physics-guided hierarchy as a baseline. Its compact representation uses local voltage, angle, frequency, and availability information; its expanded representation adds multiscale dynamics, adaptive residuals, and simultaneous PMU contrasts. A deterministic availability gate handles observed dropout. The ensemble is not DAE constrained, and its leaf votes are not treated as calibrated probabilities.

The ordinary split places new replicas of known sources in TEST. RQ4 imposes the harder contract: it removes each physical source from every source-labeled fitting and tuning step and supplies the true event family at evaluation. This isolates physical-source generalization from detection and family-classification errors.

TABLE 3. Four research questions and their evidence contracts. Results from different rows are not pooled.

RQ	Scientific question	Primary evidence and independent unit	Claim boundary
RQ1	What hidden voltage information do eight PMUs recover?	100 frozen nonlinear PowerDynamics TEST trajectories; complete trajectory is the unit	Matched model and fixed PMU set
RQ2	What fails under physical mismatch, and can a constrained recentering repair it?	980 independently rebuilt plants for diagnosis; 60 held-out equilibria for physical recentering	Point recovery under tested shifts, not sequential adaptation
RQ3	When can simultaneous load sources be resolved?	3,488 fresh nonlinear paths and 10,564 complete-record decisions over all 137 hypotheses	One event family, known onset, fixed topology and sensing
RQ4	Does supervised localization survive physical-source holdout?	303 decisions from 101 held-out event parents in ANDES	Historical baseline; true event family supplied

Figure 4 grounds the event ontology before any score is reported. A fault produces a short voltage sag, a line outage redistributes current, and injection changes evolve more slowly. Missing or corrupted measurements require different evidence. The clock trace in panel (e) makes the distinction explicit: time advances while packets from PMU 29 are absent.

All intervals and tests use the experiment's independent unit: a complete trajectory, event parent, or rebuilt plant. Generated tables and figures read only the hashed frozen snapshot. Sample-level PMU frames are never promoted to independent replicates.

VII. STATE EVIDENCE FROM RECOVERY TO MODEL FAILURE

A. RQ1 MEASUREMENT CONDITIONING RECOVERS THE TARGET

The estimator sequence reveals where the nominal accuracy comes from. Conditioning the equilibrium on the current PMU frame (B1) reduces hidden-voltage error by more than one order of magnitude. Adding causal propagation (B2) improves most trajectories, but by a much smaller amount. Thus the network-constrained measurement map supplies the main gain; temporal propagation refines it.

Figure 5(a) shows the scale change directly, and panel (c) uses paired trajectory differences rather than overlapping marginal intervals. The comparison is deliberately internal: B0, B1, and B2 share the same local physical model. It does not claim superiority over every learned or nonlinear estimator.

The structural residual in Fig. 5(b) correlates with bus-level B2 error, but the relationship is incomplete. The buses share trajectories and network coupling, and the residual comes from the threshold-sensitive projector discussed in Section IV. It is therefore a difficulty indicator, not an observability certificate.

Two negative checks sharpen this result. The fixed-lag smoother does not improve the nonlinear TEST trajectories, even though a correctly specified linear-Gaussian smoother cannot increase expected squared error. Innovation autocorrelation identifies the missing premise: the local model treats deterministic governor and load excitation as white process noise. Marginal temperature scaling repairs coverage under

nominal operation but does not whiten those innovations. More data cannot compensate for the wrong temporal model.

B. RQ2 PHYSICAL SHIFT EXPOSES THE MODEL BOUNDARY

The mismatch campaign asks whether nominal accuracy survives a changed plant. It does not fail uniformly. Network-parameter changes and operating-point shifts dominate the state error, while the tested machine, governor, AVR, and load-model perturbations remain close to the nominal level over the short excitation [Fig. 6(a)]. Those near-null responses show that the experiment did not expose the corresponding parameters; they are not evidence of general robustness.

This pattern suggests a mechanism. A stale equilibrium moves the center of both the measured and hidden voltage fields. The offset-only ablation estimates that movement from measured buses but ignores AC balance, and it closes almost none of the oracle gap. The constrained recentering in (21) instead rebuilds a physically admissible center. Across all three held-out operating shifts, it removes nearly all of the point-error gap [Fig. 7(a)–(b)].

The recovery does not identify every operating-point coordinate. The nuisance information is rank deficient, yet the requested voltage field changes little along much of that nullspace. This is the empirical counterpart of Proposition 1: a target can be recoverable even when the full latent vector is not. The remaining limitation is uncertainty. Marginal intervals cover, but the joint NEES-like statistic is severely underdispersed, and the independent constrained solver converges on only part of its small diagnostic subset. The evidence supports point reconstruction for the tested operating shifts, not joint parameter recovery or sequential adaptation.

VIII. EVENT EVIDENCE FROM DETECTION TO SOURCE AMBIGUITY

A. RQ3 CARDINALITY RESOLVES BEFORE EXACT SUPPORT

The prospective test opens the full 137-hypothesis bank only after the physical dictionary, covariance, priors, onset, and numerical integration are frozen. Every source pair competes in one normalized posterior. The resulting pattern is monotone with event strength, but the two decisions do not improve at the same rate (Table 6). Finite pairs are recovered exactly, whereas weak-weak Top-1 remains 24.4%. Weak sources are

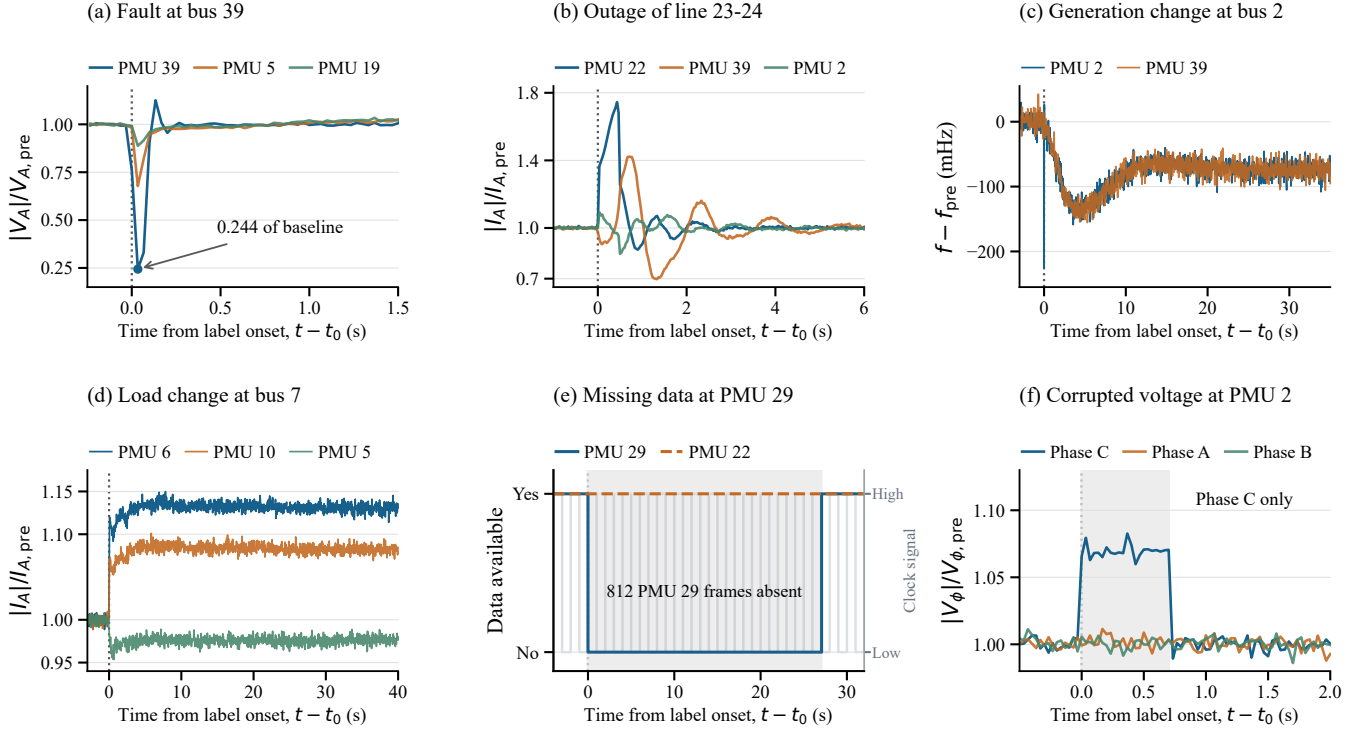


FIGURE 4. Representative measured signatures from the RAW001 reference simulation. Physical events alter voltage, current, or frequency through the network; integrity events alter availability or channel consistency. In panel (e), the gray reference clock advances while PMU 29 packets are absent. Curves are unsmoothed, and each panel uses the time scale needed to show its physical response.

TABLE 4. Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.

Method	Complex RMSE	$ V $ RMSE	Angle RMSE ($^\circ$)	Mean HVRE (%)	95% HVRE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot LMMSE	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

TABLE 5. Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

m	n	Frozen HVRE (%)	Corrected HVRE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

therefore the decisive regime, not a small perturbation around an otherwise uniform accuracy claim.

Figure 8 follows the inference failure in the same order as the method. Panel (a) establishes the strength-dependent recovery boundary. Panel (b) then shows that evidence for $K = 2$ grows while exact pair recovery remains nearly flat. The model has detected multiplicity before it has localized the support. Reporting only event detection would hide this failure.

Event presence is easier than source recovery. The frozen detector separates most event and no-event records, but its no-event exposure is too short for an operational false-alarm claim. Per-bus inclusion probabilities also rank useful candi-

dates, yet their missed-source rate remains too high to repair weak exact-support decisions. Supplementary Appendix G reports these secondary metrics and denominators.

B. PROFILED GEOMETRY EXPLAINS THE WEAK CASES

The retrospective audit asks why the weak bank fails. A support-only score compares candidate signatures while holding the rest of the explanation fixed. The profiled distance instead allows the competing support to choose its best amplitude and thereby measures the actual overlap between response manifolds. The latter tracks exact weak-case recovery far more closely [Fig. 8(c)].

This result changes the interpretation of the weak regime.

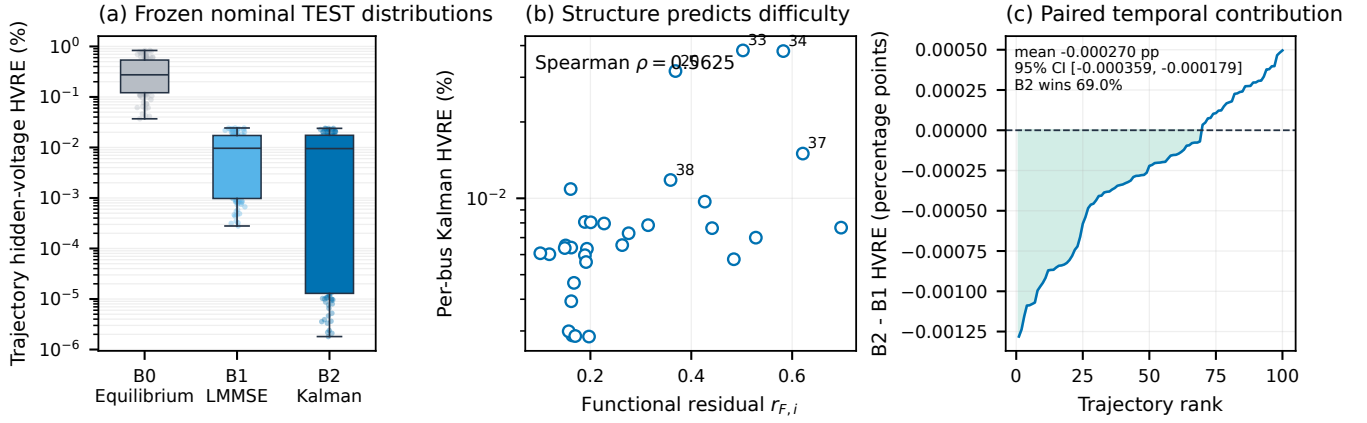


FIGURE 5. Where nominal state-reconstruction accuracy comes from. (a) PMU conditioning produces the dominant reduction in trajectory-level hidden-voltage error. (b) The preregistered structural residual predicts some, but not all, difficult buses. (c) Paired B2-minus-B1 differences isolate the smaller contribution of causal propagation.

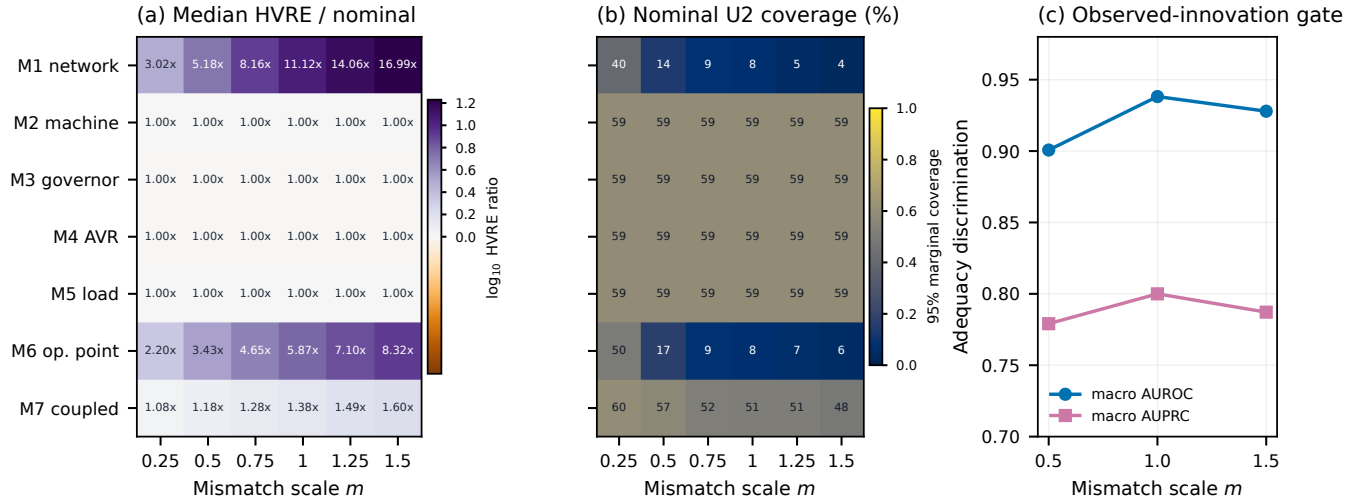


FIGURE 6. How the frozen state estimator fails under physical mismatch. (a) Network and operating-point changes dominate the hidden-voltage error. (b) A covariance correction fitted under nominal operation loses coverage after physical shift. (c) Observed innovations often flag unacceptable hidden error but do not identify its physical cause. Each family-scale cell contains 20 rebuilt plants.

Every tested direction is first-order resolvable, so a curvature-dominated a^{-4} delay law is not the observed mechanism. The errors arise because several finite-horizon candidate responses lie close after amplitude profiling. Detectability and source identity therefore require different geometric quantities.

The first-order theory separates two reasons for that proximity. A source may have little response outside the nuisance span, or it may remain visible but nearly collinear with another source after nuisance removal. Shared-source competitors make the second mechanism especially severe because their common component cancels before the exclusive source is compared. The frozen audit uses the full nonlinear profiled distance and supports this explanation retrospectively; it does not yet validate the analytic small-amplitude coefficient in (13).

The 120-frame diagnostic adds a second piece of the explanation. Extending the window increases the separation

proxy by more than eightfold, and no candidate pair is exactly degenerate in the frozen equilibrium model. Yet the hardest signatures remain almost collinear [Fig. 8(e)–(f)]. More time adds information without guaranteeing a well-conditioned decision.

The physical-manifold check also shows why the nonlinear interaction in (23) cannot be dropped. Adding only single-source curvature does not improve the simultaneous response; adding Q_{ij} reduces the median whitened error by roughly two orders of magnitude [Fig. 8(d)]. This check covers 32 nominal-operating-point paths. It validates the representation locally, not population recovery at 120 frames or robustness to operating shift.

C. RQ4 SOURCE LABELS DO NOT GENERALIZE TO NEW ASSETS

The supervised hierarchy provides a useful contrast. On the ordinary split it detects events and ranks known sources well,

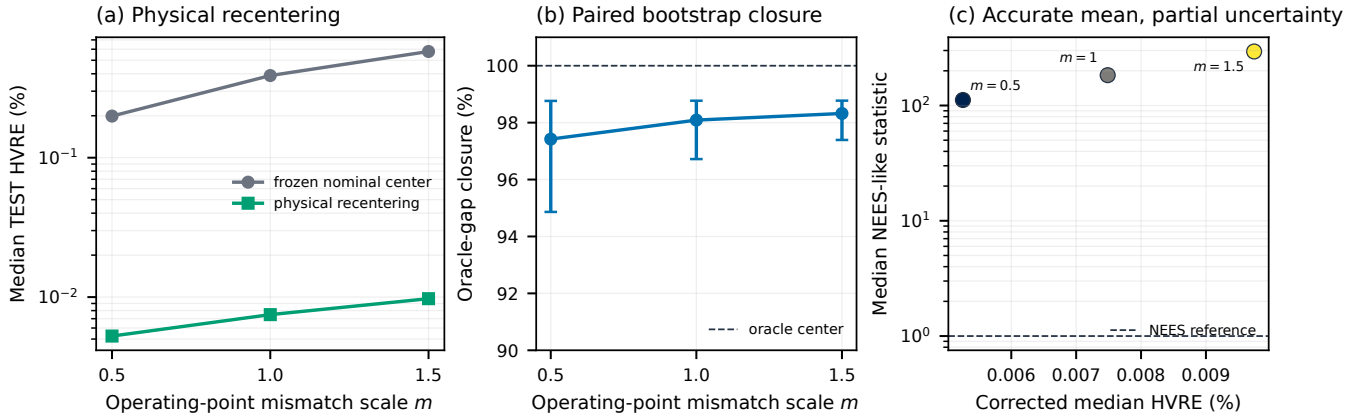


FIGURE 7. Physical recentering separates point recovery from uncertainty. (a) Enforcing the AC map sharply reduces hidden-voltage error under operating-point shift. (b) The paired oracle-gap closure is consistent across held-out cases. (c) The joint NEES-like diagnostic remains too large even when marginal intervals cover, so the point estimate is supported but its joint covariance is not.

TABLE 6. Fresh prospective 137-hypothesis confirmation. The bank contains H_0 , 16 single-source hypotheses, and all 120 two-source supports. Repeated noise realizations are decisions, not independent physical trajectories. Top- k is exact support recovery and is undefined for H_0 and single-source rows in this exported support table.

Regime	Physical trajectories	Decisions	Cardinality (%)	Top-1 (%)	Top-3 (%)	Mean $p(K = 2)$
No event	—	100	100.0	—	—	0.000
Single source	128	384	93.8	—	—	0.051
Weak-weak pair	1920	5760	37.4	24.4	59.8	0.427
Weak-strong pair	480	1440	77.6	66.2	95.8	0.808
Moderate pair	480	1440	99.7	98.8	100.0	0.996
Finite pair	480	1440	100.0	100.0	100.0	1.000

confirming that multiscale dynamics, adaptive residuals, and PMU contrasts carry event information. That result answers interpolation at represented assets.

The source-holdout result answers a different question. With the physical asset removed from all source-labeled fitting and the true event family supplied, only 3 of 303 decisions are exact. The failure remains after detection and family confusion have been removed. The location heads and asset descriptors therefore memorize source-specific associations that do not transfer to an unseen physical source.

The physical load bank avoids that shortcut by generating a response for every candidate support. Its success is narrower in another direction: it covers only load interventions under fixed topology, known onset, matched operation, and synthetic noise. Together, the two experiments motivate the next model rather than a claim of general superiority: learned residuals should correct a physical candidate likelihood and must be tested on event parents and source assets excluded from training.

IX. DISCUSSION AND LIMITATIONS

The experiments support one central distinction: recovering a requested state function is not the same as identifying the intervention that produced it. The state estimator can reconstruct hidden voltages because many unobservable state directions have little effect on that target. Source inference is stricter. It must separate every admissible candidate after

amplitude and other nuisance directions are allowed to adjust.

The state results make this distinction concrete. Instantaneous PMU conditioning supplies most of the nominal improvement, while causal propagation adds a smaller correction. When the equilibrium becomes stale, an unconstrained offset does not solve the problem; rebuilding an AC-feasible center does. The resulting point estimate is accurate even though the operating-point coordinates remain non-unique. Its joint covariance is still unreliable. Target recovery and parameter recovery must therefore be evaluated separately.

An archived ANDES calibration checkpoint makes the parameter distinction more concrete. Individual screening retained 10 of 18 directions, but their joint condition number was 722.9. Rank-revealing selection reduced the set to eight directions with condition number 7.83, and a final 6.15σ displacement gate left seven in the released local posterior. The rejected directions did not share one explanation: six were information-poor, two violated the local curvature gate, two added little first-order direction, and one left the local model domain. Supplementary Appendix D traces these counts to the frozen gate ledger. Because the checkpoint lacks the required higher derivatives, it does not establish which rejected directions are curvature-resolved or structurally redundant.

The event results expose a parallel hierarchy. Weak simultaneous events often provide enough evidence for two active sources before they identify the pair. Profiled manifold distance explains this behavior because it measures what remains

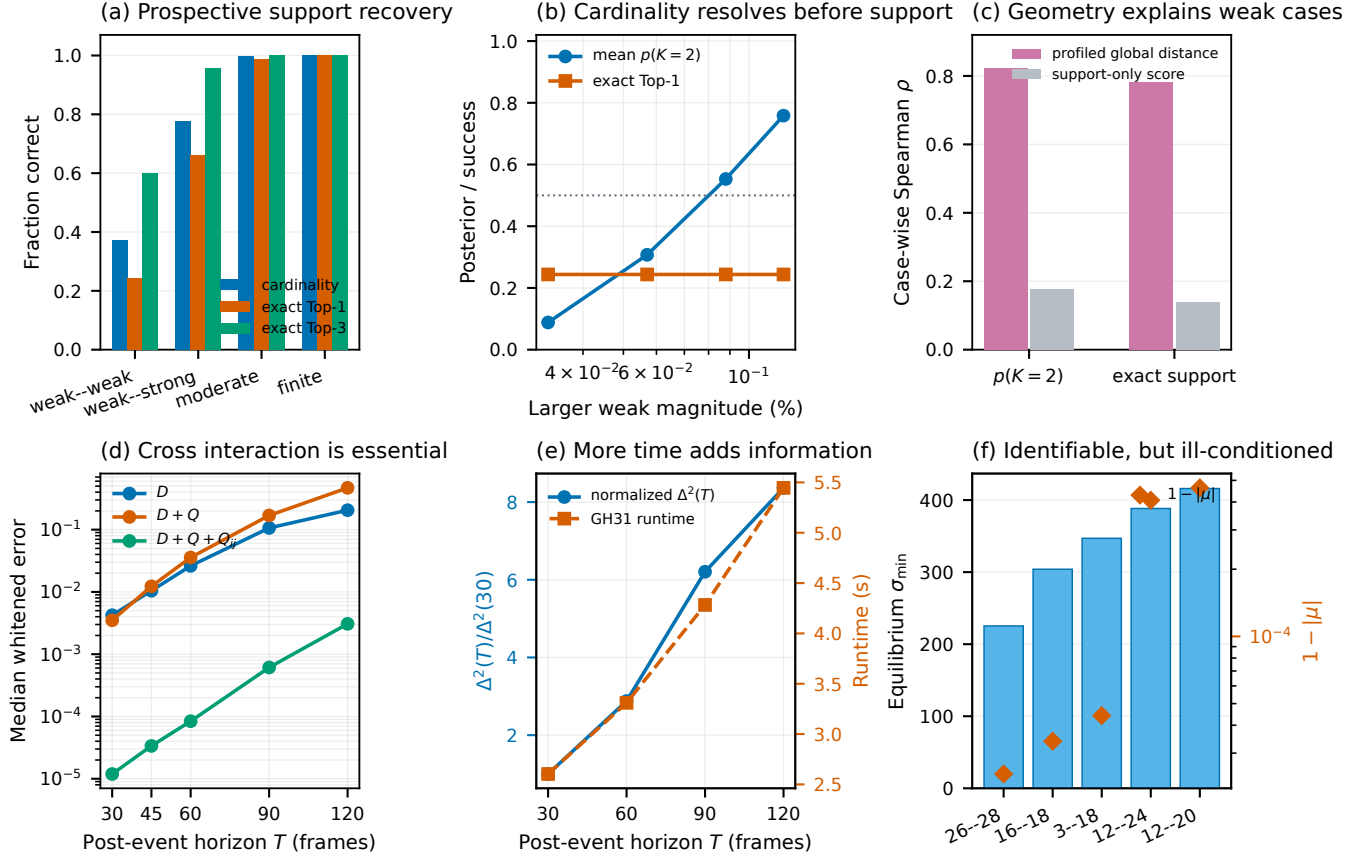


FIGURE 8. Why simultaneous-source recovery fails in the weak regime. Prospective results first separate cardinality from exact support (a–b). Retrospective geometry then links weak-case errors to profiled response distance (c). The longer-horizon diagnostic shows that the pairwise interaction is needed (d), information accumulates with time (e), and distinct equilibrium signatures can remain nearly collinear (f). Panels (c)–(f) explain the result but do not replace a prospective recovery test.

after a competing support chooses its best amplitude. Longer observation increases that distance, but nearly collinear signatures still amplify noise. A positive structural margin is therefore a uniqueness statement, not a reliability guarantee.

The supervised baseline adds a warning about learning. Engineered temporal and spatial features work when the source appears in training. Under complete source holdout, the same representation loses source identity even when the event family is supplied. A learned correction should therefore operate on candidate residuals, not on asset labels alone, and its value must be measured against the physics-only likelihood on the same event parents.

The evidence remains bounded in four ways. First, the state and event paths are evaluated separately rather than as one fitted posterior. Second, the Bayesian support result covers one event family, fixed topology, known onset, fixed PMUs, and matched operation. Third, the 120-frame study is diagnostic and lacks independent operating-point validation. Fourth, the normal exposure is too short to establish false alarms per hour, and no field or cross-simulator validation is claimed.

The next experiment should close these gaps in that order. It should first freeze the 30–120-frame recovery contract at

independent operating points. It should then compare learned-only, physics-only, residual-corrected, and diagnosability-aware variants on identical source-disjoint parents. Only after those checks should the bank expand to faults, outages, generation changes, integrity nuisance, changing PMU sets, and long continuous streams. This sequence tests the mechanism before adding scope.

X. CONCLUSION

Sparse PMUs impose different information limits on state reconstruction and event-source inference. The requested hidden-voltage function can be estimated accurately even when the full local state is not observable. Event support requires a stronger condition: candidate responses must remain separated after amplitude and nuisance directions are profiled.

The experiments follow that logic. Eight PMUs reconstruct the 31 unmeasured bus voltages with 0.009634% mean relative error under the matched model, and physical recentering repairs most of the tested operating-point shift. In the prospective load bank, finite two-source supports are recovered exactly, whereas weak-pair Top-1 remains 24.4%. The gap is explained by finite-horizon response overlap and near-

collinearity, not by event absence alone.

The present evidence supports matched-model state reconstruction and simultaneous load-source inference as separate components. A common multi-family experiment with state and integrity nuisance, source holdout, changing sensing, and physical shift is still required before a joint operational decision can be trusted.

SUPPLEMENTARY MATERIAL

The companion supplement contains the paper-by-paper literature matrix, electrical and PMU conventions, proofs, complete estimator equations, frozen experimental contracts, and secondary result tables. These details preserve reproducibility without interrupting the main scientific argument.

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Supplementary Material for Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT This supplement records the detailed literature matrix, electrical conventions, mathematical proofs, Fisher-information analysis, Bayesian estimator construction, frozen experimental contracts, and secondary results supporting the main article. It distinguishes the evaluated state and 137-hypothesis load-support estimators from the still-unexecuted common-contract joint formulation. It does not add empirical claims or change any test denominator. Every empirical table and figure is generated from the same hashed evidence snapshot used by the main manuscript; separate deterministic tables audit stated algebraic identities.

INDEX TERMS Bayesian inference, Fisher information, functional observability, phasor measurement units, reproducibility, state estimation, event-source inference.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. “Held source” means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_\theta \dot{\theta}). \quad (1)$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential states remain continuous unless a simulator-specific reset is declared.

With $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}), \quad (2)$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}). \quad (3)$$

For branch $e = (i, j)$ with complex tap t_e , series admittance y_e , and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \quad (4)$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

Positive-sequence phasors define the balanced model. Negative- and zero-sequence components remain integrity or unbalance evidence. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2 \angle V_i}{dt^2}, \quad (5)$$

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic <i>et al.</i> [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho <i>et al.</i> [2]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu <i>et al.</i> [3]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo <i>et al.</i> [4]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [5]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshtagh <i>et al.</i> [6]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey <i>et al.</i> [7]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [8]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li <i>et al.</i> [9]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow <i>et al.</i> [10]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [11]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus <i>et al.</i> [12]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai <i>et al.</i> [13]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
Main article	Hidden voltage field	All load supports with $K \leq 2$; historical eight-label ontology	DAE hypotheses and quadratic physical likelihood	Eight PMU buses	Physical bank has no source-label head	Physical/integrity variables separate

but an operational likelihood must reproduce the causal filtering of the PMU backend.

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r , hidden-voltage relative error is

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (6)$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the independent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

This appendix develops the mathematical chain used by the paper. It begins with the sampled hybrid event map, separates state-functional recovery from event-source separation, derives the relevant Fisher information after physical nuisance is removed, and then states the finite-horizon and temporal limits. The results are local unless a statement explicitly says otherwise.

A. REGULARITY AND STATISTICAL CONTRACT

Let $u = \text{col}(x, z)$ collect differential and algebraic coordinates and consider the descriptor model

$$E\dot{u} = F(u, p), \quad y = h(u, p). \quad (7)$$

The analysis uses four assumptions.

- 1) The reference angle has been fixed and the algebraic Jacobian is nonsingular on the local mode under study, so a regular index-one chart exists between discrete events.
- 2) The scheduled event time is fixed. In a neighborhood of zero amplitude, the complete callback–solver–sampling map is twice continuously differentiable.
- 3) Every covariance used as a metric is positive definite on the retained measurement subspace. Candidate-dependent covariances retain their log determinants in likelihood comparisons.
- 4) The support set is finite and all Bayesian priors are proper. Exact signed-amplitude equivalences use a

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	$\Delta P_G, \Delta Q_G$, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	$\Delta P_L, \Delta Q_L$, ZIP convention	continuous input-driven response

neighborhood of zero within each support subspace; constrained physical amplitudes instead use their actual difference set.

These assumptions distinguish the local theory from general descriptor left invertibility, state-triggered saltation, and global nonlinear identifiability.

B. SIMULATOR-CONSISTENT EVENT SENSITIVITIES

For a fixed-time, impulse-free event, the differential state is continuous, but the algebraic state must satisfy the post-event constraint. This physical right limit must be distinguished from the values stored inside a simulator callback and from the first sampled frame. The following result makes that distinction explicit.

Theorem 1 (Local consistency-jump expansion): Let $x_0 = x(\tau^-)$ and suppose $g_+(x_0, z_0, 0) = \mathbf{0}$. If g_+ is three times continuously differentiable and g_z is nonsingular at $(x_0, z_0, 0)$, there is a unique local consistent branch

$$z_+(a) = z_0 + az_1 + \frac{1}{2}a^2z_2 + O(a^3), \quad (8)$$

where all derivatives below are evaluated at the nominal right-limit mode and

$$z_1 = -g_z^{-1}g_a, \quad (9)$$

$$z_2 = -g_z^{-1}(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}). \quad (10)$$

The corresponding differential state satisfies $x(\tau^+) = x(\tau^-)$, and its first event-amplitude sensitivity starts with

$$\dot{x}_a(\tau^+) = f_a - f_z g_z^{-1}g_a \equiv B_{r,a}. \quad (11)$$

Its second-order initial forcing is

$$\begin{aligned} \dot{x}_{aa}(\tau^+) \equiv B_{r,aa} = & f_{zz}[z_1, z_1] + 2f_{za}z_1 + f_{aa} \\ & - f_z g_z^{-1}(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}). \end{aligned} \quad (12)$$

a: Proof.

The implicit-function theorem applied to $g_+(x_0, z_+(a), a) = \mathbf{0}$ gives a unique smooth local branch. Differentiating once gives $g_z z_1 + g_a = \mathbf{0}$, which yields (9). Differentiating a second time gives $g_z z_2 + g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa} = \mathbf{0}$, which yields (10). An impulse-free fixed-time switch does not reset the energy-storage coordinates, so the differential state is continuous. Differentiating $\dot{x} = f_+(x, z_+(a), a)$ at the right limit, with $x_a(\tau^+) = \mathbf{0}$, gives (11). A second differentiation, using $x_{aa}(\tau^+) = \mathbf{0}$ and (10), gives (12). $\square$

The theorem also defines an implementation-independent consistency diagnostic. If a callback changes the parameter but temporarily retains z_0 , its algebraic residual is

$$r_{cb}(a) = g_+(x_0, z_0, a) = ag_a + \frac{1}{2}a^2g_{aa} + O(a^3). \quad (13)$$

Substituting the zeroth-, first-, and second-order approximations to (8) produces constraint residuals of orders $O(a)$, $O(a^2)$, and $O(a^3)$, respectively. Central finite differences of an independently solved algebraic branch can therefore test both derivatives and the expected convergence orders without relying on undocumented callback storage.

For a differentiable PMU map $y = h(x, z, a)$, the direct right-limit signature is

$$y_a(\tau^+) = h_a - h_z g_z^{-1}g_a, \quad \Delta y_P = aM_P y_a(\tau^+) + O(a^2). \quad (14)$$

This direct algebraic visibility is not the entire event signature. Even when (14) is nearly zero at the retained PMUs, (11) can generate a distinguishable dynamic response over later frames. Instantaneous invisibility therefore does not imply finite-horizon nonidentifiability.

The frozen artifacts begin at the first retained 30-Hz frame and do not record the callback microstate or an independently solved algebraic right limit. Accordingly, the paper treats the right-limit equations above as the physical contract, not as an observed fact about the current ANDES callback implementation. The relevant differentiable object for the existing response dictionary is the sampled map

$$\Psi_{T_s}^{\text{sim}} : (u^-, p(a)) \mapsto u(\tau + T_s; a), \quad T_s = 1/30 \text{ s}. \quad (15)$$

For event coordinates a_i , define

$$L_i = \left. \frac{\partial \Psi_{T_s}^{\text{sim}}}{\partial a_i} \right|_{a=\mathbf{0}}, \quad L_{ij}^{(2)} = \left. \frac{\partial^2 \Psi_{T_s}^{\text{sim}}}{\partial a_i \partial a_j} \right|_{a=\mathbf{0}}. \quad (16)$$

For $t \geq \tau + T_s$, the first and second sensitivities obey

$$E\dot{s}_i = F_u s_i + F_p p_i, \quad (17)$$

$$E\dot{s}_{ij} = F_u s_{ij} + r_{ij}, \quad (18)$$

where

$$\begin{aligned} r_{ij} = & F_{uu}[s_i, s_j] + F_{up}[s_i, p_j] + F_{pu}[p_i, s_j] \\ & + F_{pp}[p_i, p_j] + F_p p_{ij}. \end{aligned} \quad (19)$$

The initial conditions are $s_i(\tau + T_s) = L_i$ and $s_{ij}(\tau + T_s) = L_{ij}^{(2)}$. For the affine event parameterization, $p_{ij} = \mathbf{0}$.

Theorem 2 (Propagation of event-map mismatch): Suppose two second-order models have the same nominal trajectory,

first-order sensitivities, parameter derivatives, and forcing in (18), but different second-order event-map initial conditions. Their difference $e_{ij} = s_{ij}^A - s_{ij}^B$ satisfies

$$E\dot{e}_{ij} = F_u e_{ij}, \quad e_{ij}(t) = \Phi_E(t, t_1)e_{ij}(t_1) \quad (20)$$

on the consistent solution subspace, where $t_1 = \tau + T_s$.

b: Proof.

Subtract (18) for the two models. Every forcing term cancels by assumption, leaving the homogeneous descriptor variational equation. Applying its transition operator gives (20). $\square$

When the retained PMU operator is locally affine, $y = H_y u + d$, the sampled response coefficients are

$$D_i = H_y s_i, \quad Q_i = \frac{1}{2} H_y s_{ii}, \quad Q_{ij} = H_y s_{ij}. \quad (21)$$

Thus all quadratic curvature comes from the physical solution and event map, not from PMU post-processing. This is why the dictionary must differentiate the same event semantics used to generate the trajectories.

C. EXACT FUNCTIONAL RECONSTRUCTION

Consider the finite-horizon linear experiment

$$Y = \mathcal{O}x + \varepsilon, \quad \psi = Lx, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, R), \quad R \succ \mathbf{0}. \quad (22)$$

Theorem 3 (Functional reconstruction equivalence): The following statements are equivalent: (i) $\mathcal{O}x_1 = \mathcal{O}x_2$ implies $Lx_1 = Lx_2$; (ii) $\ker \mathcal{O} \subseteq \ker L$; and (iii) there is a linear map K such that $L = K\mathcal{O}$.

a: Proof.

If (i) holds and $v \in \ker \mathcal{O}$, the observations of v and $\mathbf{0}$ coincide, hence $Lv = \mathbf{0}$ and (ii) follows. If (ii) holds, define $K(\mathcal{O}x) = Lx$ on $\text{range } \mathcal{O}$. The definition is independent of the representative because two representatives differ by a vector in $\ker \mathcal{O} \subseteq \ker L$. Extend K linearly to the observation space to obtain (iii). Finally, (iii) immediately implies (i). $\square$

Let

$$G = \mathcal{O}^T R^{-1} \mathcal{O}. \quad (23)$$

Theorem 4 (Best linear unbiased estimate of an estimable function): If the equivalent conditions of Theorem 3 hold, then

$$\hat{\psi}_* = LG^\dagger \mathcal{O}^T R^{-1} Y, \quad (24)$$

$$\text{Cov}(\hat{\psi}_*) = LG^\dagger L^T. \quad (25)$$

It has minimum covariance, in the positive-semidefinite order, among all linear unbiased estimators of Lx [14].

b: Proof.

Because $v^T G v = \|\mathcal{O}v\|_{R^{-1}}^2$, $\ker G = \ker \mathcal{O}$. Estimability gives $L = LG^\dagger G$, so taking expectation in (24) yields Lx . Direct covariance propagation gives (25). Let AY be any other linear unbiased estimator and set $T = A - LG^\dagger \mathcal{O}^T R^{-1}$.

Then $T\mathcal{O} = \mathbf{0}$ and the cross covariance vanishes because $LG^\dagger \mathcal{O}^T T^T = \mathbf{0}$. Consequently,

$$\text{Cov}(AY) = \text{Cov}(\hat{\psi}_*) + T R T^T \succeq \text{Cov}(\hat{\psi}_*). \quad (26)$$

If the kernel inclusion fails, two states induce the same data law but different targets, so no estimator can be unbiased for every state. $\square$

The pseudoinverse has this interpretation only for estimable targets. In the nonlinear DAE, a pointwise kernel test is not sufficient: local factorization requires the condition to hold on a regular constant-rank neighborhood with connected local fibers [15].

D. FISHER INFORMATION WITH PHYSICAL NUISANCE

For a differentiable Gaussian model

$$r \mid \theta \sim \mathcal{N}(\mu(\theta), \Sigma(\theta)), \quad (27)$$

the Fisher information contains both mean and covariance contributions. For scalar a ,

$$I(a) = \mu_a^T \Sigma^{-1} \mu_a + \frac{1}{2} \text{tr}(\Sigma^{-1} \Sigma_a \Sigma^{-1} \Sigma_a). \quad (28)$$

The second term must be retained when event amplitude changes the covariance. The evaluated load bank freezes Σ , so only the first term is used.

To expose nuisance geometry, consider

$$r = ad + N\nu + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad (29)$$

where ν may contain initial-state, operating-point, parameter, or integrity directions. Let $W^T W = \Sigma^{-1}$, $\bar{d} = Wd$, $\bar{N} = WN$, and

$$P_\perp = I - \bar{N}\bar{N}^\dagger. \quad (30)$$

Theorem 5 (Efficient Fisher information after nuisance profiling): For deterministic unknown ν , the efficient information for a is

$$I_{a|\nu}^{\text{det}} = \bar{d}^T P_\perp \bar{d} \quad (31)$$

$$= d^T \Sigma^{-1} d - d^T \Sigma^{-1} N (N^T \Sigma^{-1} N)^\dagger N^T \Sigma^{-1} d. \quad (32)$$

It is zero exactly when the event tangent lies in the observable nuisance span.

a: Proof.

After whitening, the score for a is $\bar{d}^T \bar{\varepsilon}$. Nuisance scores span $\{v^T \bar{\varepsilon} : v \in \text{range } \bar{N}\}$. Their Fisher inner product is Euclidean because $\mathbb{E}[\bar{\varepsilon}\bar{\varepsilon}^T] = I$. Removing the nuisance-score projection leaves $(P_\perp \bar{d})^T \bar{\varepsilon}$, whose variance is (31). Expanding the projector by an SVD gives (32). $\square$

If instead $\nu \sim \mathcal{N}(m_\nu, \Gamma)$ with a proper prior, marginalization yields

$$r \mid a \sim \mathcal{N}(ad + Nm_\nu, \Sigma + N\Gamma N^T), \quad (33)$$

$$I_{a|\nu}^{\text{Bayes}} = d^T (\Sigma + N\Gamma N^T)^{-1} d. \quad (34)$$

Woodbury's identity shows that (34) tends to the no-nuisance information as $\Gamma \rightarrow \mathbf{0}$ and to (31) as a full-rank nuisance

prior becomes diffuse. Thus zero deterministic efficient information is not a universal impossibility statement: a restrictive proper prior can still supply separation.

Corollary 1 (Pre-event information cannot reduce local event information): If pre-event data add nuisance precision $\Lambda_{\text{pre}} \succeq \mathbf{0}$, then

$$I_{a|\nu, \text{pre}} = I_{aa} - I_{a\nu}(I_{\nu\nu} + \Lambda_{\text{pre}})^{-1}I_{\nu a} \quad (35)$$

is nondecreasing in Λ_{pre} .

b: Justification.

If $\Lambda_2 \succeq \Lambda_1 \succeq \mathbf{0}$, matrix inversion reverses the order. Premultiplication and postmultiplication by the cross-information blocks preserve the positive-semidefinite order, and subtraction reverses it once more. $\square$

For a known source, known positive sign, and fixed covariance, the standardized matched statistic

$$T = \frac{d^T \Sigma^{-1} r}{\sqrt{d^T \Sigma^{-1} d}} \quad (36)$$

is standard normal under H_0 and has mean $a\sqrt{I}$ under amplitude a . A level- α one-sided test has power $1 - \Phi(z_{1-\alpha} - a\sqrt{I})$. Hence the amplitude required for power $1 - \beta$ is

$$a_{\alpha, \beta} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{I}}. \quad (37)$$

This is a conditional known-source benchmark, not the threshold of the 137-support search. Its empirical use in the paper is limited to the exploratory source-wise detectability association.

E. FIRST-ORDER MULTI-EVENT SUPPORT GEOMETRY AND ONSET INFORMATION

Let two first-order source signatures be d_g and d_j under fixed precision $G \succeq \mathbf{0}$. Allow the competing source to refit its amplitude. The local profiled distance satisfies

$$b_*(a) = a \frac{d_j^T G d_g}{d_j^T G d_j} + O(a^2), \quad (38)$$

$$\Delta_{g \rightarrow j}^2(a) = a^2 J_{g \rightarrow j} + O(a^3), \quad (39)$$

$$J_{g \rightarrow j} = d_g^T G d_g - \frac{(d_g^T G d_j)^2}{d_j^T G d_j} \geq 0. \quad (40)$$

Equation (40) is a pairwise local separation quantity, not the Fisher information for amplitudes conditional on the true support.

The pairwise formula extends to arbitrary supports. Let N_c be the unwhitened nuisance Jacobian, let $G_e = [g_1, \dots, g_M]$ collect the event tangents, and define

$$A = W N_c, \quad F = W G_e, \quad R_N = I - A A^\dagger, \quad \bar{F} = R_N F. \quad (41)$$

Theorem 6 (Profile-likelihood diagnostic quotient): Consider the whitened local experiment

$$y = A\nu + F\alpha + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I), \quad (42)$$

with deterministic unrestricted nuisance ν . Let $U_{\mathcal{D}}$ have orthonormal columns spanning $\mathcal{D} = \text{range}(\bar{F})$, and define

$$z = U_{\mathcal{D}}^T R_N y, \quad d_{\mathcal{D}} = \dim \mathcal{D} = \text{rank}(\bar{F}). \quad (43)$$

Then every profile-likelihood ratio over α depends on y only through z :

$$\min_{\nu} \|y - A\nu - F\alpha\|_2^2 = \|z - U_{\mathcal{D}}^T \bar{F}\alpha\|_2^2 + C(y), \quad (44)$$

where $C(y)$ is independent of α . The projector, quotient statistic up to an orthogonal change of coordinates, and $d_{\mathcal{D}}$ depend only on the nuisance span, not its parameterization.

a: Proof.

Profiling nuisance gives $\|R_N(y - F\alpha)\|_2^2$. Decompose this residual orthogonally into $\mathcal{D}$ and $\text{range}(R_N) \cap \mathcal{D}^\perp$. The first component is the squared norm in (44); because $\bar{F}\alpha \in \mathcal{D}$, the second is $C(y) = \|(I - U_{\mathcal{D}} U_{\mathcal{D}}^T) R_N y\|_2^2$ and contains no α . Reparameterizing nuisance without changing $\text{range}(A)$ leaves $A A^\dagger$ and therefore every stated object unchanged. $\square$

Thus $d_{\mathcal{D}}$ is the intrinsic dimension of the *linear profiled* event experiment. It is not a nonlinear global observability count and need not equal the number of PMUs. If candidates use different nuisance spans, or if nuisance has an informative proper prior, the corresponding candidate-specific profile or marginal likelihood must be retained; the common projector above is then insufficient.

Theorem 7 (Signed first-order support-to-support separation): For true support S with fixed severity direction α_S and competing support C with freely fitted local amplitudes, define

$$\gamma_{S \rightarrow C}^2(\alpha_S) = \min_{u, \beta_C} \|F_S \alpha_S - A u - F_C \beta_C\|_2^2. \quad (45)$$

Then

$$\gamma_{S \rightarrow C}^2(\alpha_S) = \|(I - P_{\bar{F}_C}) \bar{F}_S \alpha_S\|_2^2, \quad (46)$$

$$= \alpha_S^T K_{S|C} \alpha_S, \quad (47)$$

$$K_{S|C} = \bar{F}_S^T (I - P_{\bar{F}_C}) \bar{F}_S \succeq \mathbf{0}. \quad (48)$$

Here $P_{\bar{F}_C} = \bar{F}_C \bar{F}_C^\dagger$, so rank deficiency in either nuisance or event coordinates does not change the projected trajectory.

b: Proof.

For fixed β_C , minimizing (45) over u removes the component in $\text{range} A$ and leaves $\|\bar{F}_S \alpha_S - \bar{F}_C \beta_C\|_2^2$. A second least-squares projection gives (46). Symmetry and idempotence of the orthogonal projector give (47) and its positive-semidefinite order. $\square$

The theorem exposes two distinct weak-event mechanisms. For nonzero residualized single-source signatures $\bar{f}_i$ and $\bar{f}_j$,

$$\gamma_{i \rightarrow j}^2(a_i) = a_i^2 \|\bar{f}_i\|_2^2 (1 - \rho_{ij}^2), \quad \rho_{ij} = \frac{\bar{f}_i^T \bar{f}_j}{\|\bar{f}_i\|_2 \|\bar{f}_j\|_2}. \quad (49)$$

Small $\|\bar{f}_i\|$ is event–nuisance invisibility; large $|\rho_{ij}|$ is event–event aliasing after nuisance removal.

For general supports, write $H = S \cap C$, $B = S \setminus C$, and $D = C \setminus S$. The shared amplitudes can be refitted, so

$$\gamma_{S \rightarrow C}^2 = \|(I - P_{[A, F_H, F_D]}) F_B \alpha_B\|_2^2. \quad (50)$$

Thus the shared true component cancels exactly at first order for signed amplitudes or an interior parameter point. In particular, for $S = \{i, j\}$ and $C = \{i, k\}$ the relaxed margin is independent of the amplitude at i . Enlarging a competitor can only reduce the margin because its explanatory subspace grows.

The exclusive-support quotient removes the zero directions introduced by a literal support intersection. Define

$$\begin{aligned} Z_H &= [A, F_H], & R_H &= I - Z_H Z_H^\dagger, \\ \tilde{F}_B &= R_H F_B, & \tilde{F}_D &= R_H F_D. \end{aligned} \quad (51)$$

Then

$$\begin{aligned} \gamma_{S \rightarrow C}^2(\alpha_B) &= \min_{\beta_D} \|\tilde{F}_B \alpha_B - \tilde{F}_D \beta_D\|_2^2 \\ &= \alpha_B^\top K_{B|D;H} \alpha_B, \\ K_{B|D;H} &= \tilde{F}_B^\top (I - P_{\tilde{F}_D}) \tilde{F}_B \succeq 0. \end{aligned} \quad (52)$$

The first equality follows by minimizing first over the common span range Z_H and then over β_D . Moreover, $K_{B|D;H} \succ 0$ if and only if

$$\ker \tilde{F}_B = \{0\}, \quad \text{range } \tilde{F}_B \cap \text{range } \tilde{F}_D = \{0\}. \quad (53)$$

Indeed, a zero quadratic form means that $\tilde{F}_B \alpha_B$ lies in the competing range; the two conditions force $\alpha_B = 0$, and the converse is immediate. Under the coefficient normalization $\|\alpha_B\|_2 = 1$, Rayleigh–Ritz gives the worst relaxed direction and margin as

$$\min_{\|\alpha_B\|_2=1} \gamma_{S \rightarrow C}^2 = \lambda_{\min}(K_{B|D;H}). \quad (54)$$

This value depends on the physical scaling of the amplitude coordinates.

For a scale-free comparison, let Q_B, Q_D be orthonormal bases of the two conditional ranges. If $\theta_{\min}$ is their smallest principal angle,

$$\lambda_{\min}[Q_B^\top (I - Q_D Q_D^\top) Q_B] = \sin^2 \theta_{\min}. \quad (55)$$

The angle must be computed after removing nuisance and shared directions; otherwise every shared event creates a trivial zero angle. If the residual ambient dimension is $q_H = m - \text{rank } Z_H$, then $r_B + r_D > q_H$ forces a nonzero intersection between the two conditional spans. This is a span-relaxed dimensional obstruction, not yet a one-sided physical ambiguity.

If $S \subset C$ and active amplitudes may equal zero, the competitor contains the true response and $\gamma_{S \rightarrow C} = 0$. If zero is excluded but belongs to the amplitude-domain closure, the infimum remains zero. Pure best-fit geometry therefore cannot select cardinality among nested support models. The marginal likelihood in Supplementary Appendix E supplies the required amplitude-volume and support-prior terms.

F. PHYSICAL TANGENT-CONE SEPARATION

Let $\mathcal{A}_C$ be the physical amplitude domain of a competing event model at the nominal point, and let $T_C = T_{\mathcal{A}_C}(0)$ denote its Bouligand tangent cone. Signed load changes give $T_C = \mathbb{R}^{|C|}$; one-sided trip, shedding, outage-continuation, and fault-severity coordinates give nonnegative rays. Missing PMU data are a discrete measurement-mode change and are not forced into this continuous construction.

Theorem 8 (Constrained first-order separation): Suppose a competing trajectory family has a local C^2 effective chart ϕ with $\phi(0) = y_0$, full-column-rank derivative J , and local parameter cone $K = \mathbb{R}^p \times T_C$, where T_C is closed and polyhedral. Assume no distant branch approaches y_0 . For $y_*(\epsilon) = y_0 + \epsilon b + o(\epsilon)$,

$$d_C^2(\epsilon) = \epsilon^2 \text{dist}^2(b, JK) + o(\epsilon^2). \quad (56)$$

a: Proof.

For every $v \in K$, conicity gives $\phi(\epsilon v) = y_0 + \epsilon Jv + o(\epsilon)$, which yields the upper bound after taking the infimum. Conversely, an $o(\epsilon)$ -optimal local coordinate u_ϵ satisfies $\|\phi(u_\epsilon) - y_0\| = O(\epsilon)$. Full column rank and differentiability imply $\|u_\epsilon\| = O(\epsilon)$. Every convergent subsequence of u_ϵ/ϵ has a limit in the closed cone K , and division of the residual by ϵ gives the matching lower bound. The isolated-branch assumption prevents a nonlocal chart from attaining a smaller leading distance. $\square$

With A the whitened unconstrained nuisance tangent and F_C the event tangent, nuisance profiling is still exact:

$$\gamma_{\text{phys}}^2 = \min_{\beta_C \in T_C} \|R_N(b - F_C \beta_C)\|_2^2 = \text{dist}^2(R_N b, R_N F_C T_C). \quad (57)$$

The observable projection is unique even when the coefficient vector is not. Let $\mathcal{C} = R_N F_C T_C$ and $\mathcal{L} = \text{span } \mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{L}$,

$$\gamma_{\text{phys}} \geq \gamma_{\text{span}}, \quad \gamma_{\text{phys}} = \gamma_{\text{span}} \iff \Pi_{\mathcal{L}}(R_N b) \in \mathcal{C}. \quad (58)$$

Thus a positive span margin certifies physical separation, whereas a zero span margin does not prove one-sided ambiguity.

For $T_C = \mathbb{R}_+^{|C|}$, the optimum is a nonnegative least-squares projection. With residual $r = R_N(b - F_C \beta_C)$, the KKT conditions are

$$\beta_C \geq 0, \quad (R_N F_C)^\top r \leq 0, \quad \beta_j (R_N f_j)^\top r = 0. \quad (59)$$

For one true signature x and one admissible competing ray d ,

$$\gamma_{\text{ray}}^2 = \|x\|_2^2 - \frac{[d^\top x]_+^2}{\|d\|_2^2}, \quad [v]_+ = \max(0, v). \quad (60)$$

An antiparallel signature can therefore have zero span margin and a strictly positive physical margin.

For a shared one-sided support H , the exact difference coordinate satisfies $\delta_H \in T_H - \alpha_H$, which need not be a subspace. Hence the physical margin can depend on the shared

TABLE 3. Deterministic checks of conditional and one-sided geometry. These toy systems are not IEEE 39-bus evidence.

Check	Recomputed outcome
Conditional quotient	full-quotient error 0.0e+00 (pass).
Worst coefficient direction	eigenvalue-grid error 1.9e-08 (pass).
Conditional angle	naive 0.0e+00 deg; conditional 18.689 deg (pass).
Ray versus span	squared margins 0.0 versus 1.0 (pass).
One-sided shared source	squared-margin sequence 5.0, 2.0, 1.0, 1.0 (pass).
Cone enrichment	2000 trials; max violation 6.7e-16 (pass).
Nested cone	squared margin 1.4e-31 (pass).

true amplitude; the unconditional cancellation statement applies only when the relaxed solution is feasible. Competitor enrichment still cannot increase the margin whenever the old feasible set is contained in the new one. Likewise, a nested superset still has zero margin when it can match the shared amplitudes and set every added amplitude to zero.

For nontrivial closed cones $\mathcal{C}_S, \mathcal{C}_C$, the directed scale-free separation

$$\delta(\mathcal{C}_S \rightarrow \mathcal{C}_C) = \min_{\substack{x \in \mathcal{C}_S \\ \|x\|_2=1}} \text{dist}(x, \mathcal{C}_C) \quad (61)$$

vanishes exactly when the cones share a nonzero vector. It reduces to $\sin \theta_{\min}$ for subspaces, but need not be symmetric. Consequently, the classical signed-sparse condition $\text{spark}(\bar{F}) > 2K$ remains sufficient under one-sided constraints, while its failure need not imply a physically feasible ambiguity.

For a multi-source support S , the Fisher information conditional on that known support is

$$I_S(a) = J_S(a)^T G J_S(a), \quad J_S = \frac{\partial \mu_S}{\partial a_S}. \quad (62)$$

Positive definiteness of I_S permits local amplitude estimation conditional on S ; it does not exclude another support with the same mean.

Unknown onset has a separate degeneracy. For $\mu(a, \tau) = ad(\tau)$ and fixed covariance,

$$I(a, \tau) = \begin{bmatrix} d^T \Sigma^{-1} d & ad^T \Sigma^{-1} \dot{d} \\ a \dot{d}^T \Sigma^{-1} d & a^2 \dot{d}^T \Sigma^{-1} \dot{d} \end{bmatrix}. \quad (63)$$

Profiling amplitude by a Schur complement gives

$$I_{\tau|a} = a^2 \left[\dot{d}^T \Sigma^{-1} \dot{d} - \frac{(d^T \Sigma^{-1} \dot{d})^2}{d^T \Sigma^{-1} d} \right]. \quad (64)$$

At $a = 0$, onset is unidentifiable. If $\dot{d}$ is proportional to d , delay and gain are locally confounded even for nonzero amplitude. The evaluated bank avoids this unresolved problem by fixing onset.

G. CANDIDATE INDISTINGUISHABILITY AND SPARSE INJECTIVITY

For candidate h and fixed integrity support S , define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{range}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (65)$$

The profiled margin in the main paper is half the squared distance between two such sets under R_H^{-1} .

Theorem 9 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the candidate manifolds intersect. At any intersection, the two candidates induce the same Gaussian law; no decision rule can distinguish them uniformly over the declared nuisance set.

a: Proof.

Zero distance between closed affine sets means that there exist nuisance vectors for which their means coincide. The covariance is common by the stated contract, so the two conditional probability measures are identical. For any decision region, the probabilities of choosing i and j under this common law sum to one; at least one conditional error is therefore at least one half. $\square$

For two fixed simple candidates with squared Mahalanobis distance $\Delta^2 = 2D_{ij,S}$ and equal priors, projection onto their mean difference reduces the optimal test to one dimension, giving

$$P_e^* = \Phi(-\Delta/2) = \Phi(-\sqrt{D_{ij,S}/2}). \quad (66)$$

This formula does not apply directly to amplitude-marginalized support mixtures.

For the local persistent-event model $Y_T = \mathcal{H}_T a + \varepsilon_T$, define the physically admissible difference set

$$\mathcal{D}_K = \{a - b : a, b \text{ admissible and } K\text{-sparse}\}. \quad (67)$$

Theorem 10 (Finite-horizon sparse event uniqueness): Every admissible K -sparse event is uniquely determined in the noise-free local model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{D}_K = \{0\}. \quad (68)$$

For a signed neighborhood in each support subspace, $\mathcal{D}_K = \mathcal{A}_{2K}$ locally, recovering the condition in the main paper.

b: Proof.

If two admissible events produce the same output, their nonzero difference belongs to both sets in (68). Conversely, any nonzero vector in the intersection is, by definition of $\mathcal{D}_K$, the difference of two admissible K -sparse events that produce the same output. $\square$

Unknown initial-state nuisance can be removed locally by whitening and projection:

$$\tilde{\mathcal{H}}_T = (I - P_{\text{range}(W_T \mathcal{O}_T)}) W_T \mathcal{H}_T, \quad W_T^T W_T = R_T^{-1}. \quad (69)$$

Theorem 10 then applies with $\tilde{\mathcal{H}}_T$. Define the restricted margin

$$\gamma_{2K}(T) = \min_{1 \leq |U| \leq 2K} \sigma_{\min}(\tilde{\mathcal{H}}_{T,U}). \quad (70)$$

If two K -sparse explanations both fit data within whitened residual ϵ , then

$$\|a - b\|_2 \leq \frac{2\epsilon}{\gamma_{2K}(T)}. \quad (71)$$

Indeed, the triangle inequality bounds $\|\tilde{\mathcal{H}}_T(a - b)\|_2$ by 2ϵ , while the definition of γ_{2K} gives the reverse lower bound. For the evaluated $K \leq 2$ bank, the relevant union size is four.

H. INFORMATION CANNOT DECREASE WHEN VALID MEASUREMENTS ARE ADDED

For a true trajectory at severity a and a competing physical family with parameter $\eta \in \Theta_C$, define the exact profiled prefix distance

$$d_T^2(a) = \inf_{\eta \in \Theta_C} r_T(a, \eta)^\top \Sigma_T^{-1} r_T(a, \eta), \quad (72)$$

where every Σ_T is a principal prefix of one positive-definite process covariance and the physical competitor domain does not change with T . Partition the difference between candidate means into old and added blocks, $\delta = \text{col}(\delta_1, \delta_2)$, with joint covariance $\Sigma \succ 0$. Block Gaussian elimination gives

$$\delta^\top \Sigma^{-1} \delta = \delta_1^\top \Sigma_{11}^{-1} \delta_1 + \tilde{\delta}_2^\top \Sigma_{2|1}^{-1} \tilde{\delta}_2, \quad (73)$$

$$\tilde{\delta}_2 = \delta_2 - \Sigma_{21} \Sigma_{11}^{-1} \delta_1, \quad (74)$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}. \quad (75)$$

Theorem 11 (Prefix monotonicity and temporal rescue): Under the prefix and fixed-domain contract above,

$$d_{T+1}^2(a) \geq d_T^2(a). \quad (76)$$

If both minima are attained, equality holds if and only if an element of $\text{Argmin } d_T^2$ also has zero conditional innovation residual in the added block. Suppose further that, locally,

$$d_T^2(a) = c_T a^{2k_T} + o(a^{2k_T}), \quad c_T > 0. \quad (77)$$

Then $k_{T+1} \leq k_T$; if the order remains k , then $c_{T+1} \geq c_T$.

a: Proof.

The conditional covariance in (73) is positive definite, so its second quadratic form is nonnegative for every fixed η . Taking the infimum over the same Θ_C proves (76). If the minima are attained, equality of the infima forces an extended-horizon minimizer to minimize the prefix term and to make the innovation term zero; the converse is immediate. If $k_{T+1} > k_T$, the ratio $d_{T+1}^2(a)/d_T^2(a)$ tends to zero as $a \rightarrow 0$, contradicting (76). When both orders equal k , division by a^{2k} and passage to the limit yield $c_{T+1} \geq c_T$. $\square$

The theorem permits a longer prefix to reveal a first-order direction that was absorbed at an earlier horizon, or to reveal a higher-order obstruction that was previously invisible. It does not say that a concrete estimator improves: changing its normalization, nuisance set, covariance model, or optimization breaks the prefix contract. In particular, the archived RAWSIM39 calibration renormalized each event-signal-horizon block to unit weight and is not a prefix likelihood for this theorem.

If whitened model errors under two supports satisfy $\|We_S\| \leq \epsilon_S$ and $\|We_R\| \leq \epsilon_R$, the reverse triangle inequality gives the robust separation bound

$$d_{\text{true}} \geq [d_{\text{nom}} - \epsilon_S - \epsilon_R]_+, \quad \mathcal{J}_{\text{true}} \geq \frac{1}{2} [d_{\text{nom}} - \epsilon_S - \epsilon_R]_+^2. \quad (78)$$

This is the required bridge for future simulator-shift claims; it is not evaluated by the current nominal load bank.

I. TEMPORAL INFORMATION UNDER THE FROZEN AR(1) CONTRACT

For true support S , competing support R , true amplitude a , and competitor amplitude b , let $\delta_t(b) = \mu_{S,t}(a) - \mu_{R,t}(b)$. The frozen temporal covariance is

$$\Sigma_T = R_T(\rho) \otimes \Omega, \quad [R_T]_{ij} = \rho^{|i-j|}, \quad |\rho| < 1. \quad (79)$$

For fixed b , its squared Mahalanobis separation has the innovation form

$$q_T(b) = \|\delta_1(b)\|_{\Omega^{-1}}^2 + \frac{1}{1 - \rho^2} \sum_{t=2}^T \|\delta_t(b) - \rho \delta_{t-1}(b)\|_{\Omega^{-1}}^2. \quad (80)$$

Define the profiled simple-model separation

$$\mathcal{J}_{S \rightarrow R}(T) = \frac{1}{2} \inf_{b \in \mathcal{A}_R} q_T(b). \quad (81)$$

It equals the minimum KL divergence over fixed-amplitude simple Gaussian competitors. It is not the KL divergence or Bayes factor of the amplitude-mixture support model.

Theorem 12 (Persistent AR(1) information rate): If the amplitude domain is compact and fixed and $\delta_t(b)$ converges uniformly to $\delta_\infty(b)$, then

$$\lim_{T \rightarrow \infty} \frac{2\mathcal{J}_{S \rightarrow R}(T)}{T} = \frac{1 - \rho}{1 + \rho} \inf_{b \in \mathcal{A}_R} \|\delta_\infty(b)\|_{\Omega^{-1}}^2. \quad (82)$$

a: Proof.

Under uniform convergence, $\delta_t - \rho \delta_{t-1}$ converges uniformly to $(1 - \rho)\delta_\infty$. Divide (80) by T ; the first-frame term vanishes and the Cesàro mean converges uniformly to $(1 - \rho)^2(1 - \rho^2)^{-1} \|\delta_\infty\|_{\Omega^{-1}}^2$. Compactness and uniform convergence permit interchange of infimum and limit, giving (82). $\square$

Monotonicity of $\mathcal{J}_{S \rightarrow R}(T)$ follows independently from Theorem 11; the AR(1) calculation supplies the explicit limiting rate.

If the profiled equilibrium distance is positive, separation grows as $\Theta(T)$. If an admissible competitor matches equilibrium and its innovation differences are square summable, evaluating (81) at that competitor gives a finite upper bound; monotonicity then implies convergence to a finite ceiling. Exact trajectory overlap gives zero separation for every horizon.

For a true pair $\{i, j\}$, a competing pair $\{i, k\}$ can refit the shared source. With whitened equilibrium columns $\bar{h}_i, \bar{h}_j, \bar{h}_k$,

$$a_j^2 \|(I - P_{i,k})\bar{h}_j\|_2^2 \leq a_j^2 \|(I - P_i)\bar{h}_j\|_2^2. \quad (83)$$

The inequality follows from $\text{span}(\bar{h}_i) \subseteq \text{span}(\bar{h}_i, \bar{h}_k)$. It formalizes why identifying which weak second source occurred cannot be easier than rejecting the shared single-source explanation.

J. HIGHER-ORDER CONTACT AND LOCAL DETECTION SCALE

First-order Fisher information can vanish while nonlinear separation remains. In whitened coordinates, let the no-event nuisance manifold and event curve be

$$F_0(b) = Nb + \frac{1}{2}H_0[b, b] + O(\|b\|^3), \quad (84)$$

$$c(a) = ad + a^2q + O(a^3). \quad (85)$$

Theorem 13 (Relative first- and second-order separation): Assume the nuisance manifold has constant rank and use a regular chart after removing redundant coordinates. Define the profiled first-order coefficient

$$\gamma_1^2 = \|P_\perp d\|_2^2, \quad P_\perp = I - (WJ_c)(WJ_c)^\dagger, \quad (86)$$

where J_c is the nuisance tangent before whitening and d is the already whitened event tangent in (85). If $\gamma_1 > 0$, the local squared distance to the nuisance manifold is

$$\Delta^2(a) = a^2\gamma_1^2 + O(a^3). \quad (87)$$

If $\gamma_1 = 0$, so the whitened event tangent lies in the nuisance tangent space, choose the regular-chart vector v satisfying $Nv = d$ and define

$$\kappa = P_\perp \left(q - \frac{1}{2}H_0[v, v] \right). \quad (88)$$

Then

$$\Delta^2(a) = a^4\gamma_2^2 + O(a^5), \quad \gamma_2^2 = \|\kappa\|_2^2. \quad (89)$$

a: Proof.

The local minimizer satisfies $DF_0(b)^\top[F_0(b) - c(a)] = 0$. At the origin, its derivative with respect to the regular chart coordinate is $N^\top N \succ 0$, so the implicit-function theorem gives a smooth minimizing branch. Write $b_*(a) = av + a^2w + O(a^3)$. At first order, $N^\top(Nv - d) = 0$, yielding residual $aP_\perp d + O(a^2)$ and (87). If $d = Nv$, the first-order residual vanishes. At second order, minimization projects $q - \frac{1}{2}H_0[v, v]$ onto range N ; the remaining normal component is (88). Squaring the residual gives (89). The projector is unchanged by any full-rank reparameterization of the same nuisance tangent space, so γ_1 is a property of the local manifolds and covariance metric rather than of a particular nuisance coordinate basis. $\square$

The same constant-rank qualification makes γ_2 intrinsic. Redundant original parameters may give several representatives of the same observable tangent, but their normal acceleration equals the second fundamental form of the same manifold branch. If $\gamma_2 = 0$, the conclusion is only $\Delta^2(a) = o(a^4)$ under the corresponding little- o expansion; an a^6 law or exact overlap requires a separate third-order analysis.

K. OPERATIONAL ERROR AND SUPPORT-RECOVERY LIMITS

For two simple hypotheses with a shared covariance, $P_i = \mathcal{N}(\mu_i, \Sigma)$, define

$$d^2 = (\mu_1 - \mu_0)^\top \Sigma^{-1} (\mu_1 - \mu_0). \quad (90)$$

The equal-prior log likelihood ratio is Gaussian with mean $\mp d^2/2$ and variance d^2 under P_0 and P_1 , respectively. Hence

$$P_e^* = \Phi(-d/2), \quad D_{\text{KL}}(P_1 \| P_0) = d^2/2, \quad (91)$$

$$C_{\text{Ch}} = d^2/8.$$

The last identity gives $P_e^* \leq \frac{1}{2} \exp(-d^2/8)$. If a common affine nuisance is $A\eta$, orthogonal projection after whitening removes it exactly and replaces d by the quotient margin. Thus $P_e^* = \Phi(-\gamma/2)$ is exact for that quotient Gaussian experiment. It is not an exact formula for two nonlinear amplitude mixtures with different priors or dimensions.

Suppose the first nonzero contact order at horizon T is k and

$$d_T^2(a) = T\mathcal{I}_k |a|^{2k} [1 + o(1)], \quad \mathcal{I}_k > 0. \quad (92)$$

For target simple-Gaussian error $\alpha < 1/2$, inversion of (91) yields

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}} \right]^{1/k}. \quad (93)$$

Equivalently, if $a_T = cT^{-\beta}$, the optimal binary error tends to one half for $\beta > 1/(2k)$, to a nontrivial constant at equality, and to zero for $\beta < 1/(2k)$. This gives the $T^{-1/2}$ and $T^{-1/4}$ boundaries for first- and second-order separation.

Now let $\Theta = \bigcup_{S \in \mathcal{S}} \Theta_S$ and define the minimax exact-support risk

$$R^* = \inf_{\hat{S}} \sup_{\theta \in \Theta} \Pr\{\hat{S} \neq S(\theta)\}. \quad (94)$$

Theorem 14 (Binary and minimum-severity limits): For every pair $\theta_i \in \Theta_{S_i}$ with $S_0 \neq S_1$,

$$R^* \geq \frac{1}{2} [1 - \text{TV}(P_{\theta_0}, P_{\theta_1})] \geq \frac{1}{2} \left[1 - \sqrt{\frac{1}{2} D_{\text{KL}}(P_{\theta_0} \| P_{\theta_1})} \right]_+. \quad (95)$$

If $S^+ = S \cup \{j\}$ and $P_{S^+, a_j} \rightarrow P_S$ in total variation as $a_j \downarrow 0$, then uniform exact-support recovery over all $a_j > 0$ has minimax risk at least one half.

a: Proof.

Restriction of any support rule to a two-point subset cannot reduce its worst-case risk; the optimal equal-prior binary risk is $[1 - \text{TV}(P, Q)]/2$. Pinsker's inequality gives the second bound. For the nested pair, the binary risk tends to one half as the active amplitude tends to zero, so the supremum over positive amplitudes is at least that limit. $\square$

The theorem requires either a declared $|a_j| \geq a_{\min} > 0$ contract or a decision that may abstain near absence. It also distinguishes a source-ranking error from an impossible exact-cardinality decision.

For M equiprobable supports, Fano's inequality gives

$$P_e \geq 1 - \frac{I(V; Y) + \log 2}{\log M}. \quad (96)$$

With $\bar{P} = M^{-1} \sum_j P_j$, convexity of KL in its second argument implies

$$\begin{aligned} I(V; Y) &= \frac{1}{M} \sum_i D_{\text{KL}}(P_i \| \bar{P}) \\ &\leq \frac{1}{M^2} \sum_{i,j} D_{\text{KL}}(P_i \| P_j) = \bar{D}_{\text{KL}}. \end{aligned} \quad (97)$$

Therefore a necessary condition for $P_e \leq \alpha$ is

$$\bar{D}_{\text{KL}} \geq (1 - \alpha) \log M - \log 2. \quad (98)$$

If $M = \binom{N}{K}$ and $\bar{D}_{\text{KL}}(T) \leq T \bar{\mathcal{I}}_k a^{2k}/2$, then necessarily

$$T \geq \frac{2[(1 - \alpha) \log \binom{N}{K} - \log 2]}{\bar{\mathcal{I}}_k a^{2k}}. \quad (99)$$

This is an impossibility certificate, not a sufficient sample-size guarantee [16], [17]. For sparse $K \ll N$, it has the familiar scaling $T \gtrsim K \log(N/K)/(\bar{\mathcal{I}}_k a^{2k})$, now expressed through the physical Hybrid-DAE information rate.

L. DIAGNOSABILITY-PRESERVING DISCREPANCY COVARIANCE

A residual correction can improve fit while making physical events harder to distinguish. The following calculation makes that tradeoff explicit. Work after whitening and removal of common nuisance. Let $Q_U \in \mathbb{R}^{m \times r}$ be an orthonormal basis for the discriminant directions that must be protected, and let $Q_D \in \mathbb{R}^{m \times q}$ be an orthonormal discrepancy basis. The former should be frozen from critical candidate pairs; the latter must be estimated from mismatch-only records so that unlabeled events are not absorbed as model error.

Theorem 15 (Discrepancy retention and finite intensity): Let $P_D = Q_D Q_D^T$ and model the whitened residual with covariance

$$C_s = I + sP_D, \quad s = \tau^2 \geq 0. \quad (100)$$

Then $C_s^{-1} = I - \alpha P_D$, where $\alpha = s/(1 + s)$, and the worst information fraction retained over the protected subspace is

$$\rho(s, Q_D) = 1 - \alpha \sigma_{\max}^2(Q_U^T Q_D). \quad (101)$$

Set $c^2 = \sigma_{\max}^2(Q_U^T Q_D)$ and $\delta = 1 - \rho_{\min}$. The constraint $\rho \geq \rho_{\min}$ imposes no finite upper bound on s when $c^2 \leq \delta$; otherwise it requires

$$s \leq s_{\max} = \frac{\delta}{c^2 - \delta}. \quad (102)$$

If the true residual covariance is $I + C_m$ and $E_D = \text{tr}(Q_D^T C_m Q_D)$, the expected Gaussian negative twice-log-likelihood relative to $s = 0$, up to constants, is

$$\Delta \mathcal{L}(s) = q \log(1 + s) - \frac{s}{1 + s} (q + E_D). \quad (103)$$

For fixed Q_D , its constrained minimizer is finite:

$$s^* = \min\{E_D/q, s_{\max}\}, \quad (104)$$

with $s_{\max} = +\infty$ in the first geometric case.

TABLE 4. Deterministic checks for the discrepancy-retention identities. These are algebraic tests, not power-system experiments.

Check	Error	Tolerance	Status
Low-rank covariance inverse	3.34e-16	1.0e-12	PASS
Principal-angle retention	3.33e-16	1.0e-12	PASS
Retention variance cap	0.00e+00	1.0e-12	PASS
Finite full-likelihood intensity	5.50e-09	1.0e-07	PASS
Safe projected PCA	8.88e-16	1.0e-10	PASS
Protected-distance floor	1.11e-16	1.0e-12	PASS

a: Proof.

The inverse follows from $P_D^2 = P_D$. For $x = Q_U a$, the information ratio is the Rayleigh quotient of $I - \alpha Q_U^T P_D Q_U$; its smallest eigenvalue is (101) because the singular values of $Q_U^T Q_D$ are the principal-angle cosines. Solving $\alpha c^2 \leq \delta$ gives (102). Finally, $\log |C_s| = q \log(1 + s)$ and taking the expectation of $y^T C_s^{-1} y$ under $\text{Cov}(y) = I + C_m$ gives (103). Its derivative is $(qs - E_D)/(1 + s)^2$, which gives (104). $\square$

Under strict no-loss protection, $\rho_{\min} = 1$ and $s > 0$ require $\text{range}(Q_D) \subseteq \text{range}(Q_U)^\perp$. For fixed rank, the basis that captures the most mismatch energy is therefore the top- q eigenspace of $P_{U^\perp} C_m P_{U^\perp}$, provided that the orthogonal complement has dimension at least q . This is a safe projected-PCA baseline, not a learned-estimator result.

Maximizing only αE_D omits both the covariance determinant and the whitened measurement-noise contribution. It therefore drives s toward an infinite-variance limit whenever geometry permits. That utility is a valid description of reduced quadratic mismatch penalty, but it is not a normalized Bayesian tuning objective. The general joint orientation problem for $q > 1$ is an offline nonconvex design; spectral, semidefinite, or moment relaxations belong in a separate optimization study unless they are executed on the frozen event and mismatch dictionaries.

M. APPLIED GATE TAXONOMY FROM AN ARCHIVED ANDES CHECKPOINT

The frozen RAWSIM39 V1.2 calibration ledger provides an applied boundary check on the theory. It contains complete first-order screening and rank-selection decisions for 18 plant/event directions, but not the Jacobian and higher-order arrays needed to evaluate γ_2 or γ_3 . Table 5 therefore classifies only what the recorded gates establish. In particular, failure of the curvature gate is not relabeled as second-order identifiability, and a small orthogonal fraction is not relabeled as exact redundancy.

N. THEORY-TO-EVIDENCE BOUNDARY

Equation (87) supplies a direct falsification test: for independently generated nonlinear trajectories with decreasing $|a|$, the profiled squared distance divided by a^2 should converge to γ_1^2 . That convergence study has not been run for the present ANDES bank; the retrospective distance association and the gate-only RAWSIM39 taxonomy are not substitutes for it.

If n independent Gaussian replications have local distance $\Delta(a) = |a|^r \|k_r\| + O(|a|^{r+1})$, their KL divergence to the

TABLE 5. Conservative taxonomy of the frozen RAWSIM39 V1.2 calibration checkpoint. The gate sequence was 18 candidates, 10 individually eligible ($\kappa = 722.91$), eight rank-selected ($\kappa = 7.83$), and seven released after one 6.15σ local-domain rejection.

Category	Count	Directions	Evidence boundary
Regular local	7	plant_inertia_scale, plant_governor_droop_scale, generation_delta_p_pu, generation_delta_q_pu, generation_amplitude, generation_peak_s, load_delta_p_fraction	Released only under the local restricted posterior
Weak first-order	6	plant_damping, plant_governor_lag_scale, plant_turbine_time_scale, plant_avr_gain_scale, plant_avr_time_scale, plant_subtransient_x_scale	Low information; order and temporal rescue unresolved
Curvature unresolved	2	fault_duration_s, line_duration_s	Linear surrogate failed; not evidence of order 2
Near-redundant first-order	2	generation_tau_s, generation_residual	Small new orthogonal fraction; exact redundancy unproved
Local-domain failure	1	fault_reactance_pu	MAP left the declared local prior domain

TABLE 6. Recomputed checks of higher-order evidence and information limits. Controlled systems are not IEEE 39-bus performance evidence.

Check	Recomputed outcome
Evidence classes	slopes 0.9994, 0.4951, and 0.0003 for regular, order-2, and redundant additions (pass).
Prefix monotonicity	500 correlated-noise trials; max violation 0.0e+00 (pass).
Temporal rescue	first-order $\gamma_1^2(2) = 0.452724$; second-order relative error $1.0e-04$ (pass).
Gaussian error map	quadrature error $5.6e-17$ (pass).
Fano lower bound	$M = 8$ bounds 0.5993/0.3974; no Monte Carlo violation (pass).

TABLE 7. Deterministic checks of the local separation identities. These toy systems are not IEEE 39-bus evidence.

Check	Prediction and recomputed outcome
Regular first-order DAE	slope 2 and $\gamma_1^2 = 0.4527244641$; 2.000000; relative error $3.8e-12$ (pass).
Shared-tangent second order	slope 4 and $\gamma_2^2 = 2.89$; 4.000000; coefficient 2.890000 (pass).
Rank-deficient nuisance	unique projected fit and $\gamma_1^2 = 0.5$; rank 2/3; max variation $1.6e-16$ (pass).
Signed shared-source cancellation	margin independent of shared amplitude; squared-margin range $0.0e+00$ (pass).
Competitor enrichment	$\gamma(S \rightarrow C_2) \leq \gamma(S \rightarrow C_1)$; 0.0196 versus 0.5096 (pass).
Nested support	geometric margin 0; $6.1e-35$ (pass).
Principal angle	unit margin $\sin \theta$; absolute error $0.0e+00$ (pass).
Nuisance enlargement	profiled margin cannot increase; sequence 2.0, 1.0, 0.0 (pass).
Diagnostic quotient	profiled costs factor through d_D coordinates; rank 4/6; max error $4.4e-16$ (pass).
Nuisance reparameterization	unchanged nuisance span gives the same projector; spectral error $7.1e-16$ (pass).

closest null is $n\Delta^2(a)/2$. Pinsker's inequality bounds the power of any level- α test by

$$\text{Power}_a \leq \alpha + \min\{1, \sqrt{n}\Delta(a)/2\}. \quad (105)$$

Thus $n|a_n|^{2r} \rightarrow 0$ prevents power from separating from size. The reference scales are $n^{-1/2}$ for first-order contact and $n^{-1/4}$ for second-order contact; singular-information asymptotics are established more broadly [18]. Correlated frames in one PMU window are not independent replications and must be handled through their joint covariance.

All audited load directions in the present bank have nonzero first-order separation. The second-order result is therefore a boundary theorem, not the empirical explanation for weak-pair failure.

TABLE 8. Open checks for the simulator event semantics.

Check	Evidence required for closure
Right limit	Solver-scale $\ g_+(x(\tau + \epsilon), z(\tau + \epsilon), a)\ $ after re-equilibration; the callback residual may remain $O(a)$.
Algebraic derivatives	Agreement of analytic z_1, z_2 with central differences on a decreasing symmetric amplitude grid.
Taylor order	Constraint and state-approximation slopes 1, 2, 3, or amplitude-halving ratios 2, 4, 8.
Event families	One declared semantics and tolerance set for load, generation, and continuous branch-admittance perturbations.
Dynamic first order	Convergence of $\Delta y/a$ to $B_{r,a}$ over the declared early-time window.
Dynamic second order	Analytic second variation within a preregistered tolerance of nonlinear trajectories; the current discrepancy is unresolved.

TABLE 9. Status of the mathematical results relative to the frozen evidence.

Result	Present evidence boundary
Consistency jump	DAE contract only; callback/right-limit logging and derivative convergence remain open.
Functional equivalence and BLUE	Exact in the local linear model; nonlinear TEST accuracy remains empirical.
Fisher, support, and cone geometry	Signed and one-sided toy identities pass; IEEE 39-bus convergence and physical cone margins remain untested.
Sparse injectivity and γ_{2K}	T120 diagnostics concern the frozen load bank, not general descriptor left-invertibility.
Temporal order and rate	Prefix and controlled correlated-noise identities pass; the frozen long-horizon load study remains diagnostic and lacks independent operating-point validation.
Relative curvature	The controlled parabola passes; curvature-dominated weak-event behavior was not observed in the frozen atlas.
Error and minimax limits	Gaussian quotient, minimum-severity, and Fano formulas pass controlled checks; no IEEE 39 pairwise oracle-error atlas has been frozen.
RAWSIM39 gate taxonomy	The 18-direction ledger supports five conservative gate categories; derivative arrays needed to assign higher-order classes are unavailable.
Discrepancy retention	Exact for the whitened low-rank covariance model and checked algebraically; the protected dictionary, mismatch covariance, learned correction, and shifted IEEE 39-bus test remain unexecuted.

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

This appendix separates two evaluated computations from the unexecuted common estimator. Sections E-A–E-C specify the state recursion and load-support posterior that produced the reported results. The complete statistical object is the candidate-conditioned joint posterior and its normalized evidence; filtering, smoothing, or optimization is the numerical route used to approximate it. Section E-F specifies that common state–event–integrity object, but its final integration has

not yet been evaluated.

A. BAYESIAN RECURSION AND THE STATE HIERARCHY

For a Markov state model, Bayesian filtering alternates prediction and correction [19]. More broadly, this construction treats the PMU task as a Bayesian inverse problem in which physical candidate models and latent states are compared through predictive evidence [20]:

$$p(x_k | y_{0:k-1}) = \int p(x_k | x_{k-1})p(x_{k-1} | y_{0:k-1}) dx_{k-1}, \quad (106)$$

$$p(x_k | y_{0:k}) = \frac{p(y_k | x_k)p(x_k | y_{0:k-1})}{\int p(y_k | x_k)p(x_k | y_{0:k-1}) dx_k}. \quad (107)$$

The denominator in (107) is the predictive evidence. Omitting it is harmless for a state mean under one fixed model, but not when physical candidates are compared.

B0 returns the equilibrium. B1 conditions the equilibrium prior on one PMU frame:

$$K_{\text{LMMSE}} = P_0 C^T (C P_0 C^T + R)^{-1}, \quad (108)$$

$$\hat{\delta x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0). \quad (109)$$

B2 retains state and covariance between frames. For the linear Gaussian model, substituting Gaussian densities into (106)–(107) yields

$$m_k^- = A_d m_{k-1}^+, \quad P_k^- = A_d P_{k-1}^+ A_d^T + Q, \quad (110)$$

$$S_k = C P_k^- C^T + R, \quad K_k = P_k^- C^T S_k^{-1}, \quad (111)$$

$$m_k^+ = m_k^- + K_k(y_k - C m_k^-), \quad P_k^+ = (I - K_k C) P_k^-. \quad (112)$$

Theorem 16 (Linear-Gaussian reduction): Equations (110)–(112) are the exact posterior mean and covariance generated by the Bayesian recursion for the stated linear Gaussian model.

a: Proof.

An affine transformation of the Gaussian posterior at $k - 1$, followed by addition of independent Gaussian process noise, gives (110). The pair (x_k, y_k) is jointly Gaussian with cross covariance $P_k^- C^T$ and measurement covariance S_k . The standard conditional-Gaussian formula gives (111)–(112). Equivalently, multiplying the predictive density by the likelihood and completing the square gives posterior precision $(P_k^-)^{-1} + C^T R^{-1} C$; Woodbury's identity produces the same covariance and mean. $\square$

The implementation may use the Joseph covariance form

$$P_k^+ = (I - K_k C) P_k^- (I - K_k C)^T + K_k R K_k^T \quad (113)$$

to preserve symmetry and positive semidefiniteness numerically.

B3 uses the Rauch–Tung–Striebel backward recursion [21]:

$$J_t = P_t^+ A_d^T (P_{t+1}^-)^{-1}, \quad (114)$$

$$m_{t|k} = m_t^+ + J_t(m_{t+1|k} - m_{t+1}^-), \quad (115)$$

$$P_{t|k} = P_t^+ + J_t(P_{t+1|k} - P_{t+1}^-)J_t^T. \quad (116)$$

It estimates a past state using later measurements. The negative B3 result therefore diagnoses model mismatch; it does not imply that correct future information generally lacks value.

The nominal uncertainty correction rescales the rectangular voltage covariance as

$$\Sigma_{i,k}^{\text{U2}} = T \Sigma_{i,k} T^T, \quad T = \text{diag}(\sqrt{\tau_{\mathcal{R}}}, \sqrt{\tau_{\mathcal{S}}}), \quad (117)$$

with temperatures fitted on independent calibration trajectories. The observed-only adequacy statistic is a five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^T \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (118)$$

Neither operation changes the physical state estimate.

B. CONSTRAINED PHYSICAL RECENTERING

The static recentering estimator treats active/reactive load changes and active generator changes as nuisance coordinates d . It solves

$$(\hat{V}, \hat{d}) = \arg \min_{V,d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{\text{AC}}(V, d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2. \quad (119)$$

Development data fix $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. Because the AC constraint appears through a finite penalty rather than a normalized stochastic density, the implemented objective is a regularized physical estimator, not a fully normalized posterior.

For an exactly constrained local MAP, let $r(\chi)$ be the stacked residual, $c(\chi) = \mathbf{0}$ the equality constraints, and Ω the residual precision. A Gauss–Newton step solves

$$\begin{bmatrix} H & G_c^T \\ G_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \chi \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla J \\ -c \end{bmatrix}, \quad H = J_r^T \Omega J_r + H_{\text{prior}}. \quad (120)$$

Theorem 17 (Feasible local covariance): If G_c has full row rank and Z spans $\ker G_c$, then $Z^T H Z \succ \mathbf{0}$ makes the KKT matrix nonsingular. The local constrained Gaussian covariance is

$$P_\chi = Z(Z^T H Z)^{-1} Z^T. \quad (121)$$

a: Proof.

Any vector in the KKT nullspace satisfies $G_c v = \mathbf{0}$, hence $v = Zq$. Premultiplying the first block equation by v^T gives $q^T Z^T H Z q = 0$, so $q = \mathbf{0}$ and $v = \mathbf{0}$. Full row rank then forces the multiplier to zero. Parameterizing feasible perturbations by $\delta \chi = Zq$ gives Gaussian precision $Z^T H Z$ in q ; its linear pushforward is (121). $\square$

This covariance is local and unimodal. The reported joint NEES-like failure shows that it is not calibrated for the full shifted-plant experiment.

C. EVALUATED PHYSICAL LOAD-SUPPORT MODEL

Let r be the centered T -frame PMU window. For support S with amplitudes a_S , the evaluated mean is

$$\mu_S(a_S) = \sum_{i \in S} (a_i D_i + a_i^2 Q_i) + \sum_{\substack{i, j \in S \\ i < j}} a_i a_j Q_{ij}. \quad (122)$$

The W2 likelihood is

$$q_S(a_S) = \|r - \mu_S(a_S)\|_{\Sigma_T^{-1}}^2, \quad (123)$$

$$p(r | S, a_S) = \frac{\exp[-q_S(a_S)/2]}{(2\pi)^{n_T/2} |\Sigma_T|^{1/2}}, \quad (124)$$

$$\Sigma_T = R_T(\rho) \otimes \Omega. \quad (125)$$

The covariance is fitted only on normal calibration records and remains fixed across supports. The selected priors are

$$p(K = 0, 1, 2) = (0.20, 0.50, 0.30), \quad (126)$$

$$p(S | K) = \binom{16}{K}^{-1}, \quad (127)$$

$$a_i | i \in S \sim \mathcal{N}(0, \sigma_a^2), \quad \sigma_a = 0.05. \quad (128)$$

For reference, the linear single-source model has a closed form.

Proposition 1 (Conjugate amplitude posterior and Bayes factor): Let $r = ad + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \Sigma)$, and $a \sim \mathcal{N}(0, \sigma_a^2)$. Define $I = d^T \Sigma^{-1} d$ and $s = d^T \Sigma^{-1} r$. Then

$$v = (I + \sigma_a^{-2})^{-1}, \quad m = vs, \quad (129)$$

$$a | r, S \sim \mathcal{N}(m, v), \quad (130)$$

$$\log \frac{p(r | S)}{p(r | H_0)} = \frac{1}{2} \frac{s^2}{I + \sigma_a^{-2}} - \frac{1}{2} \log(1 + \sigma_a^2 I). \quad (131)$$

a: Proof.

Terms in the joint log density that depend on a are $-\frac{1}{2}[(I + \sigma_a^{-2})a^2 - 2sa]$. Completing the square gives mean m and variance v . Integrating the resulting Gaussian contributes $\sqrt{v/\sigma_a^2} \exp(vs^2/2)$ relative to H_0 ; taking its logarithm gives (131). The logarithmic term is the amplitude-volume or Occam penalty. $\square$

Corollary 2 (One-sided amplitude evidence): Under Proposition 1, replace the signed Gaussian slab by the proper half-normal prior $p(a) = 2\mathcal{N}(a; 0, \sigma_a^2)\mathbb{I}(a \geq 0)$. If $BF_{\pm}$ denotes the signed Bayes factor in (131), then

$$BF_+ = 2\Phi(z)BF_{\pm}, \quad z = \frac{\sigma_a s}{\sqrt{1 + \sigma_a^2 I}}. \quad (132)$$

Conditional on the active model, the severity posterior is $\mathcal{N}(m, v)$ truncated to $[0, \infty)$.

b: Proof.

The fixed-amplitude likelihood ratio is $\exp(as - Ia^2/2)$. Multiplication by the half-normal density and completion of the square give twice the signed Gaussian integral restricted to $a \geq 0$. The unrestricted posterior has standardized probability $\Phi(m/\sqrt{v}) = \Phi(z)$ on that half-line, which yields (132). $\square$

The factor $2\Phi(z)$ preserves direction: a large score with the prohibited sign is evidence against the one-sided event even though the signed factor depends on s^2 . Proper slab scales are essential because Bayes factors across different support dimensions are undefined up to an arbitrary constant under an improper amplitude prior. Under the one-sided null $a = 0$, the likelihood ratio statistic satisfies

$$2\Delta\ell = [Z]_+^2, \quad Z \sim \mathcal{N}(0, 1), \quad (133)$$

and therefore has the boundary mixture $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$, rather than the regular χ_1^2 law.

The quadratic multi-source mean is nonconjugate. Its evidence is

$$Z_S = p(r | S) = \int p(r | S, a_S) \prod_{i \in S} p(a_i) da_S. \quad (134)$$

With Q -node Gauss–Hermite abscissas x_q and weights w_q , the Gaussian-prior transformation $a_i = \sqrt{2}\sigma_a x_{q_i}$ gives

$$Z_S \approx \pi^{-|S|/2} \sum_{q_1=1}^Q \cdots \sum_{q_{|S|=1}^Q \left(\prod_{j=1}^{|S|} w_{q_j} \right) p(r | S, \sqrt{2}\sigma_a x_q). \quad (135)$$

The implementation uses $Q = 31$ and evaluates (135) in the log domain using log-sum-exp. Independent adaptive Gauss–Kronrod integration checks representative two-dimensional cases. These checks support numerical integration under the frozen model; they do not validate the physical model itself.

D. POSTERIOR OUTPUTS AND THE CARDINALITY-LOCALIZATION DISTINCTION

The normalized support posterior is

$$p(S | r) = \frac{Z_S p(S)}{\sum_{R \in \mathcal{H}} Z_R p(R)}. \quad (136)$$

It induces cardinality, source-inclusion, and amplitude distributions:

$$p(K = k | r) = \sum_{|S|=k} p(S | r), \quad (137)$$

$$p(i \in S | r) = \sum_{S: i \in S} p(S | r), \quad (138)$$

$$p(a_i | r) = p(i \notin S | r)\delta_0 + \sum_{S: i \in S} p(S | r)p(a_i | S, r). \quad (139)$$

Equation (139) is a model-averaged spike-and-slab output: source absence contributes a point mass at zero.

The experimentally observed gap between cardinality and support has an exact algebraic decomposition. Let $N_k = \binom{16}{k}$,

$\ell_S = \log Z_S$, and $\ell_k^* = \max_{|S|=k} \ell_S$. Define the effective multiplicity

$$m_{\text{eff},k} = \sum_{|S|=k} \exp(\ell_S - \ell_k^*). \quad (140)$$

Proposition 2 (Exact effective-multiplicity identity): Under a uniform prior over supports of cardinality k ,

$$\log p(r | K = k) = \ell_k^* - \log N_k + \log m_{\text{eff},k}, \quad (141)$$

$$\max_{|S|=k} p(S | K = k, r) = m_{\text{eff},k}^{-1}. \quad (142)$$

a: Proof.

Factor $\exp(\ell_k^*)$ from $N_k^{-1} \sum_{|S|=k} \exp(\ell_S)$ and take logarithms to obtain (141). Conditional normalization within cardinality k divides the largest evidence by the same sum, giving (142). $\square$

Many similar pair supports can therefore increase evidence for $K = 2$ while preventing any one pair from dominating. Entropy separates the same two questions exactly because K is a deterministic function of S :

$$H(S | r) = H(K | r) + \sum_k p(K = k | r) H(S | K = k, r). \quad (143)$$

The first term measures cardinality uncertainty; the second measures localization ambiguity within each cardinality.

For set-valued output, sort supports by posterior mass and define the smallest credible set

$$C_q(r) = \arg \min_{\mathcal{C}} |\mathcal{C}| \quad \text{subject to} \quad \sum_{S \in \mathcal{C}} p(S | r) \geq q. \quad (144)$$

Candidate-set size and coverage are more informative than an arbitrary Top- m when the posterior is multimodal. The present paper has not yet calibrated an abstention threshold or credible-set coverage prospectively.

E. STEP-BY-STEP EVALUATED BAYESIAN ALGORITHM

Table 10 records the exact order of the load-support computation. Steps marked “evaluated” produced the 137-hypothesis results; optional outputs are mathematically defined but are not promoted to validated claims.

The direct quadrature cost per record is

$$1 + GQ + \binom{G}{2} Q^2 \quad (145)$$

likelihood-node evaluations for $K \leq 2$. With $G = 16$ and $Q = 31$, this is 115,817 nodes before vectorization or reuse. The count explains why the support bank, integration order, and horizon must be frozen together when runtime is compared.

F. CANDIDATE-CONDITIONED JOINT STATE-EVENT DESIGN

The common-contract extension would infer state trajectory X , algebraic variables Z , event support S , amplitudes a_S ,

operating nuisance η , and integrity corruption b . Its posterior factorization is

$$p(X, Z, \eta, b, S, a_S | Y) \propto p(S) p(a_S | S) p(\chi_0) \\ \times \prod_{t=0}^{T-1} p(\chi_{t+1} | \chi_t, S, a_S) \prod_{t=0}^T p(y_t | \chi_t, b_t, S, a_S), \quad (146)$$

subject to the candidate DAE constraints. Here χ_t denotes a consistent state/nuisance chart. A constrained negative twice-log density, excluding $p(S)$, is

$$\mathcal{J}_S^o = \|\chi_0 - \bar{\chi}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^T \|r_{S,t}^{\text{pmu}}\|_{R_t^{-1}}^2 \\ + \sum_{t=0}^{T-1} \|r_{S,t}^{\text{dyn}}\|_{Q_t^{-1}}^2 + 2\lambda_b \sum_{t=0}^T \|b_t\|_1. \quad (147)$$

Candidate evidence must be evaluated in a common feasible measure. The boundary correction is exposed by an explicit concentration parameter rather than by informally assuming a concentrated posterior.

Theorem 18 (Local cone-Laplace evidence): Let $L_{S,\epsilon}$ be the log likelihood plus the log of a proper prior in identifiable local coordinates of dimension d_S . Suppose the rescaled feasible domains converge locally to a closed polyhedral cone K_S and

$$L_{S,\epsilon}(\xi_0 + \epsilon u) - L_{S,\epsilon}(\xi_0) = h_S^T u - \frac{1}{2} u^T H_S u + r_\epsilon(u), \quad (148)$$

where $H_S \succ \mathbf{0}$ and $r_\epsilon \rightarrow 0$ uniformly on compact sets. Assume an integrable dominating envelope and negligible evidence from distant modes. Then

$$Z_{S,\epsilon} = e^{L_{S,\epsilon}(\xi_0)} \epsilon^{d_S} (2\pi)^{d_S/2} |H_S|^{-1/2} \\ \times \exp\left(\frac{1}{2} h_S^T H_S^{-1} h_S\right) \Pr(U_S \in K_S) [1 + o(1)], \quad (149)$$

$$U_S \sim \mathcal{N}(H_S^{-1} h_S, H_S^{-1}).$$

a: Proof.

Set $\xi = \xi_0 + \epsilon u$, which contributes the Jacobian ϵ^{d_S} . The compact-uniform expansion gives pointwise convergence of the rescaled integrand to $\exp(h_S^T u - u^T H_S u/2)$. The assumed envelope permits dominated convergence, while the distant-mode condition localizes the original integral. Completing the square in the limiting quadratic integral gives the Gaussian normalizer and the cone probability in (149). $\square$

For an interior model $K_S = \mathbb{R}^{d_S}$ and the probability factor is one. For one or more boundary amplitudes it is a Gaussian orthant probability, not generally 2^{-q} . Half-normal prior normalizers supply their own factors in addition to this cone probability. Chart Jacobians, prior normalizers, and candidate-dependent covariance determinants belong in $L_{S,\epsilon}$ or the measure term. Comparing only optimized residuals drops both volume and boundary terms and does not produce a posterior.

TABLE 10. Evaluated 137-hypothesis Bayesian load-support algorithm.

Step	Operation	Computation and frozen choice	Status
0	Declare contract	Fix 16 candidate load buses, $K \leq 2$, known onset, 30-frame window, eight PMUs, nominal topology, and no corrupted channels	Evaluated
1	Validate plant	Check equilibrium, terminal-current orientation, sampling, nonlinear solve acceptance, and first-order response before building the bank	Evaluated
2	Build physical dictionary	Generate centered D_i , self-curvature Q_i , and cross-interaction Q_{ij} from simulator-consistent interventions	Evaluated
3	Fit noise on CAL	Estimate channel covariance and AR(1) dependence from normal calibration records; freeze W2 before event TEST	Evaluated
4	Freeze priors and integrator	Set $p(K)$, uniform support priors, Gaussian amplitude slab, GH31 nodes/weights, and independent GK checks	Evaluated
5	Form the residual	Align the known onset, subtract the frozen normal mean, retain the fixed PMU/channel order, and apply the declared temporal covariance	Evaluated
6	Enumerate supports	Construct H_0 , 16 single-source supports, and 120 two-source supports; no learned source label prunes the bank	Evaluated
7	Integrate each support	Evaluate (134) with (135); retain quadrature-weighted amplitude moments	Evaluated
8	Normalize once	Add the support prior in log space, subtract the global log-sum-exp, and obtain (136)	Evaluated
9	Read decisions in order	Report event presence, $p(K r)$, conditional support ranking, source inclusion, and conditional amplitude intervals separately	Evaluated
10	Quantify ambiguity	Compute entropy, effective multiplicity, or C_q without relabeling them as validated abstention guarantees	Defined; partial evidence
11	Record provenance	Store the plant identifier, support, amplitude, noise seed, likelihood version, and frozen artifact hash for every independent record	Evaluated

For a small-noise Gaussian Hybrid-DAE trajectory map $\mu_S(\xi_0 + \epsilon u) = \mu_0 + \epsilon J_S u + O(\epsilon^2)$,

$$h_S = J_S^T \Sigma^{-1} z, \quad H_S = J_S^T \Sigma^{-1} J_S, \quad (150)$$

so positive curvature is equivalent to first-order identifiability in the reduced coordinates. This is a theoretical local specialization, not a validation of the unevaluated joint estimator. If H_S is singular, merely replacing its inverse by a pseudoinverse is invalid: redundant coordinates must first be quotiented out, while genuine higher-order identification changes the evidence scale. For example,

$$\int_0^\infty e^{-na^4/4} da = \frac{\Gamma(1/4)}{4^{3/4}} n^{-1/4}, \quad (151)$$

not the $n^{-1/2}$ rate of regular quadratic Laplace theory.

Proposition 3 (Exact reduction along redundant fibers): Suppose a parameter chart $\theta \leftrightarrow (\xi, \zeta)$ is one-to-one on the prior support and the observable map satisfies $\mu(\theta) = \bar{\mu}(\xi)$ there. If $\tilde{\pi}(\xi, \zeta)$ is the transformed proper prior, then the evidence equals the reduced integral with induced prior

$$\bar{\pi}(\xi) = \int \tilde{\pi}(\xi, \zeta) d\zeta. \quad (152)$$

Thus an exactly redundant coordinate contributes no vanishing small-noise power; its effect is the pushforward prior measure, not a pseudodeterminant.

b: Proof.

The likelihood is independent of ζ . Fubini's theorem integrates the proper transformed prior along each fiber and gives $Z = \int p(y | \xi) \bar{\pi}(\xi) d\xi$ exactly. $\square$

For a merely local chart, the same reduction is asymptotic only after a localization condition excludes evidence from distant branches.

Theorem 19 (Mixed regular and higher-order evidence): In whitened coordinates, suppose the first observable expansion after redundancy removal is

$$\mu(\xi, a) = \mu_0 + A\xi + \frac{c_k}{k!} a^k + o(\|\xi\| + |a|^k), \quad (153)$$

where A has rank r , the proper prior density is continuous and positive at the origin, and

$$c_{k,\perp} = (I - P_A)c_k \neq 0. \quad (154)$$

Assume a dominating envelope and negligible evidence away from this branch. For the Gaussian-normalizer-free evidence kernel with noise scale ϵ ,

$$M_\epsilon \sim C \epsilon^{r+1/k}, \quad (155)$$

$$C = \pi(0, 0) (2\pi)^{r/2} |A^T A|^{-1/2} \quad (156)$$

$$\times \frac{\Gamma(1/(2k))}{k} \left[\frac{2(k!)^2}{\|c_{k,\perp}\|_2^2} \right]^{1/(2k)}. \quad (157)$$

c: Proof.

Use $\xi = \epsilon u$ and $a = \epsilon^{1/k} t$. The Jacobian is $\epsilon^{r+1/k}$ and the limiting kernel is $\exp[-\|Au + c_k t^k/k!\|^2/2]$. The component of c_k in range A disappears after a translation of u . Gaussian integration over the remaining regular coordinates yields $(2\pi)^{r/2} |A^T A|^{-1/2}$; integration of the normal component over $t \in \mathbb{R}$ gives the gamma-function factor. Dominated convergence supplies the prior density at the origin and the stated limit. $\square$

For one additional amplitude, the theorem separates three cardinality cases. A regular direction supplies an $O(\epsilon)$ Occam factor, a direction first visible at order k supplies $O(\epsilon^{1/k})$, and an exactly redundant extension supplies an $O(1)$ factor determined by its induced prior. For $\epsilon \asymp n^{-1/2}$ these become $n^{-1/2}$, $n^{-1/(2k)}$, and n^0 . These are local evidence rates; the RAWSIM39 gate ledger does not contain the higher-order derivatives required to assign them to its rejected parameters.

For a sliding window, eliminating old Gaussian variables creates an arrival prior. Write its canonical log kernel as $-\frac{1}{2} z^T \Lambda z + h^T z + c$. If the information matrix and vector are partitioned into variables a to eliminate and retained variables

TABLE 11. Deterministic checks of boundary-aware evidence identities. These toy systems are not evaluated estimator results.

Check	Recomputed outcome
Signed/one-sided Bayes factors	max quadrature error 2.7e-12 (pass).
Broad nuisance limit	relative information error 8.9e-10 (pass).
Boundary likelihood ratio	zero mass 0.4972; positive mean 0.9987 (pass).
Nonlinear cone-Laplace	relative-error slope 0.9497 (pass).
Singular quartic evidence	scaling slope -0.2500 (pass).

b , exact integration yields

$$\Lambda_{\text{new}} = \Lambda_{bb} - \Lambda_{ba}\Lambda_{aa}^{-1}\Lambda_{ab}, \quad (158)$$

$$h_{\text{new}} = h_b - \Lambda_{ba}\Lambda_{aa}^{-1}h_a, \quad (159)$$

$$c_{\text{new}} = c + \frac{1}{2}h_a^T\Lambda_{aa}^{-1}h_a + \frac{n_a}{2}\log(2\pi) - \frac{1}{2}\log\det\Lambda_{aa}. \quad (160)$$

The first two lines preserve the next MAP problem; the last line is also required to preserve evidence between candidates.

Table 12 gives the intended sequence. It is a specification for the next experiment, not a claim about the results reported in the main paper.

For a posterior state $\pi_t(S) = p(S | Y_{1:t})$, an idealized unit-cost support decision with per-frame delay cost c satisfies the Bellman equation

$$V(\pi) = \min \left\{ 1 - \max_S \pi(S), c + \mathbb{E}[V(\pi_{t+1}) | \pi_t = \pi] \right\}. \quad (161)$$

The one-step expected information gain is

$$I(S; y_{t+1} | Y_{1:t}) = H(S | Y_{1:t}) - \mathbb{E}[H(S | Y_{1:t+1})]. \quad (162)$$

Equations (161)–(162) motivate adaptive horizon selection. They are not an executed stopping rule for the current fixed-onset, fixed-window bank.

Finally, the hidden-voltage posterior must average across support uncertainty rather than report only the winning support. With $w_S = p(S | Y)$, conditional target mean m_S , and covariance P_S ,

$$\bar{m} = \sum_S w_S m_S, \quad (163)$$

$$\text{Cov}(\psi | Y) = \sum_S w_S P_S + \sum_S w_S (m_S - \bar{m})(m_S - \bar{m})^T. \quad (164)$$

The second term is between-support ambiguity. Dropping it would make the virtual-PMU field appear overconfident precisely when source hypotheses disagree.

G. ESTIMATOR CLAIM BOUNDARY

The executed state recursion is Bayesian under its local linear Gaussian model, and the load-source bank computes normalized Bayesian evidence under the frozen W2/L2 model. The physical recentering is a regularized constrained estimator with separately assessed uncertainty. The full candidate-conditioned state–event–integrity posterior, learned discrepancy, unknown-onset inference, changing-PMU operation, and sequential stopping rule remain specified but unexecuted.

No result in the main paper is attributed to those pending components.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

TABLE 12. Candidate-conditioned joint estimator design and current implementation boundary.

Step	Operation	Required calculation	Present status
1	Pre-event update	Use only pre-event measurements to update state and operating nuisance; retain cross covariance and normalized arrival evidence	State filter evaluated; joint nuisance pending
2	Integrity mask	Mark missing channels exactly and introduce sparse corruption variables for available but suspect channels	Ontology defined; common inference pending
3	Candidate generation	Enumerate physical family, support, onset, and severity domain without using source identity as a learned label	Load family evaluated; other families pending
4	Candidate prediction	Apply the candidate DAE/event map and the current PMU membership to generate a physical trajectory	Load bank evaluated
5	Candidate-conditioned solve	Minimize (147) on the feasible DAE manifold using the KKT system and continuation from the pre-event state	Not executed jointly
6	Learned discrepancy	If used, predict only candidate-residual discrepancy and covariance; do not replace the physical hypothesis or consume target source labels	Not implemented
7	Evidence normalization	Retain log determinants, prior densities, feasible-coordinate measures, and the Schur normalizer before comparing candidates	GH31 load evidence evaluated; joint Laplace evidence pending
8	Posterior outputs	Return state-functional mixture, cardinality, support, integrity state, severity, and candidate-set uncertainty	Separate state/load outputs only
9	Sequential decision	Continue observing when expected information gain justifies delay; use distinct thresholds for detection, cardinality, and exact source	Not evaluated
10	Reconfigure and verify	After accepting a physical change, rebuild the operating center and check subsequent innovations before declaring a persistent model update	Static recentering only

TABLE 13. Complete frozen-contract inventory. The main paper groups these contracts by research question.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh H_0 + 16 singles + 120 pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument. The two compact tables first preserve the exact mismatch and weak-signal values; the full-width panels then show the corresponding temporal, likelihood, and supervised-baseline diagnostics.

TABLE 14. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

TABLE 15. Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.

$ a $ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

TABLE 16. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

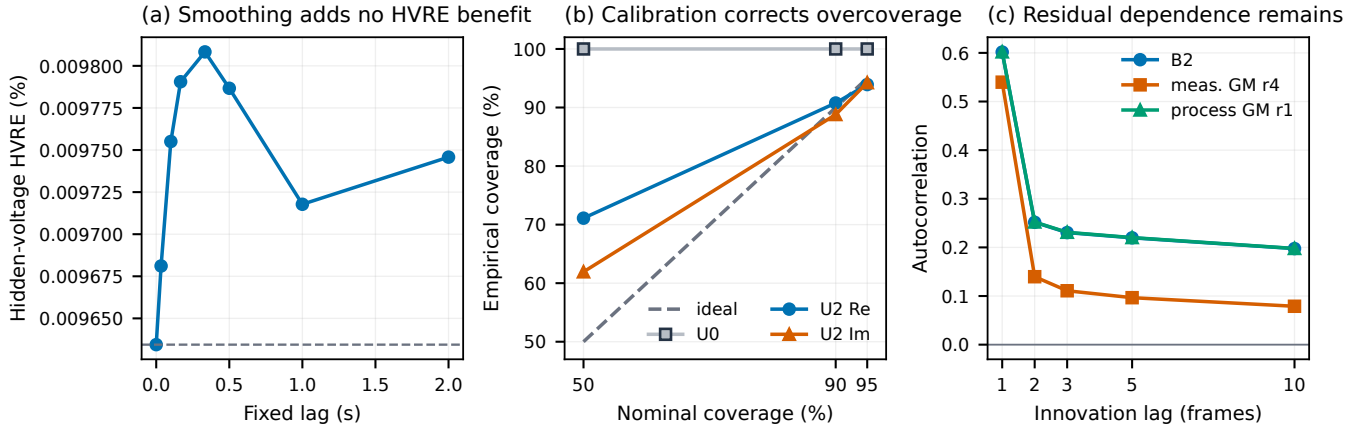


FIGURE 1. Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.

TABLE 17. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

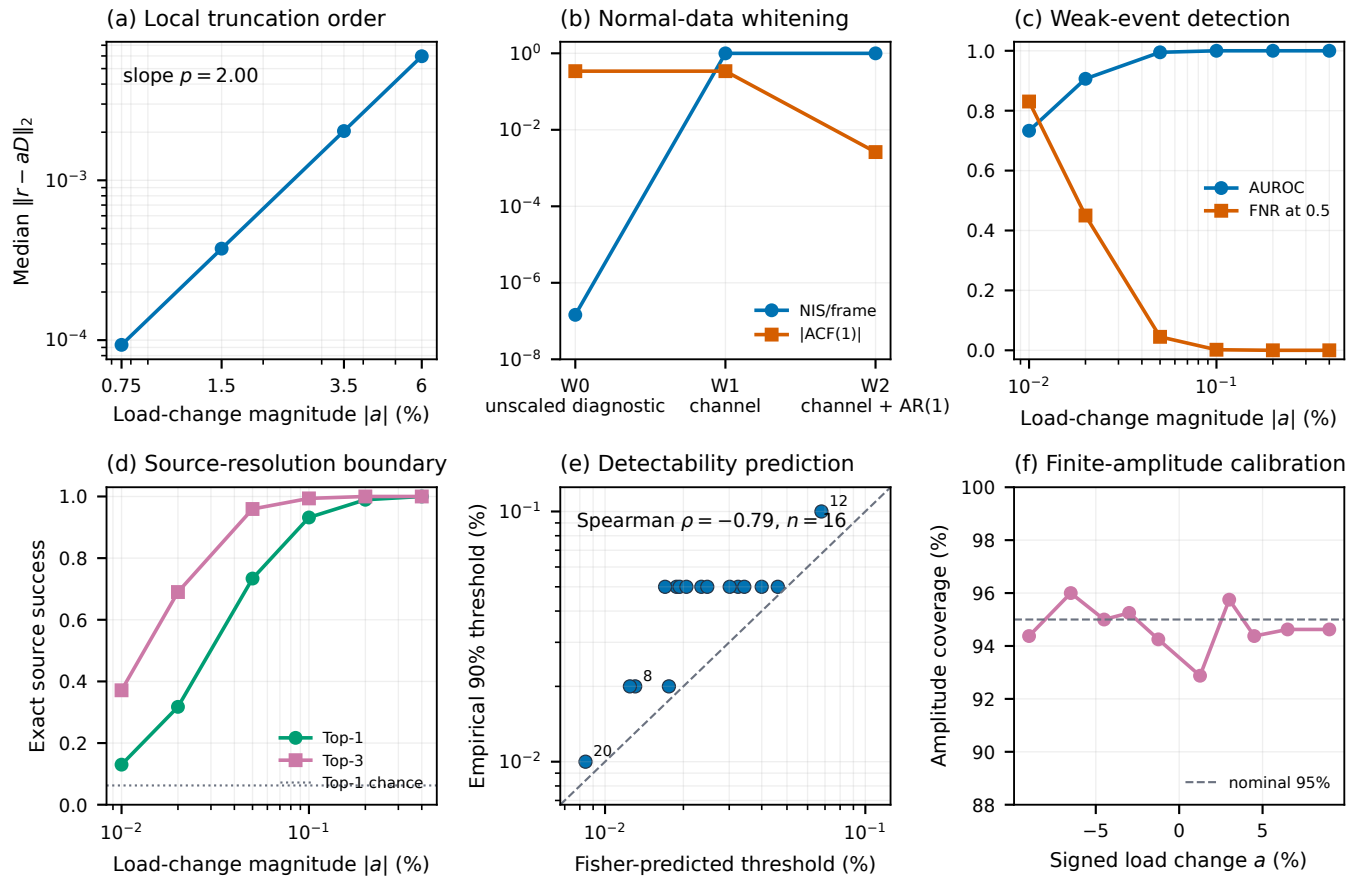


FIGURE 2. Single-source development evidence for the frozen load likelihood. The figure is retained as component validation; GLOBAL-137 in the main paper is the prospective support-recovery result.

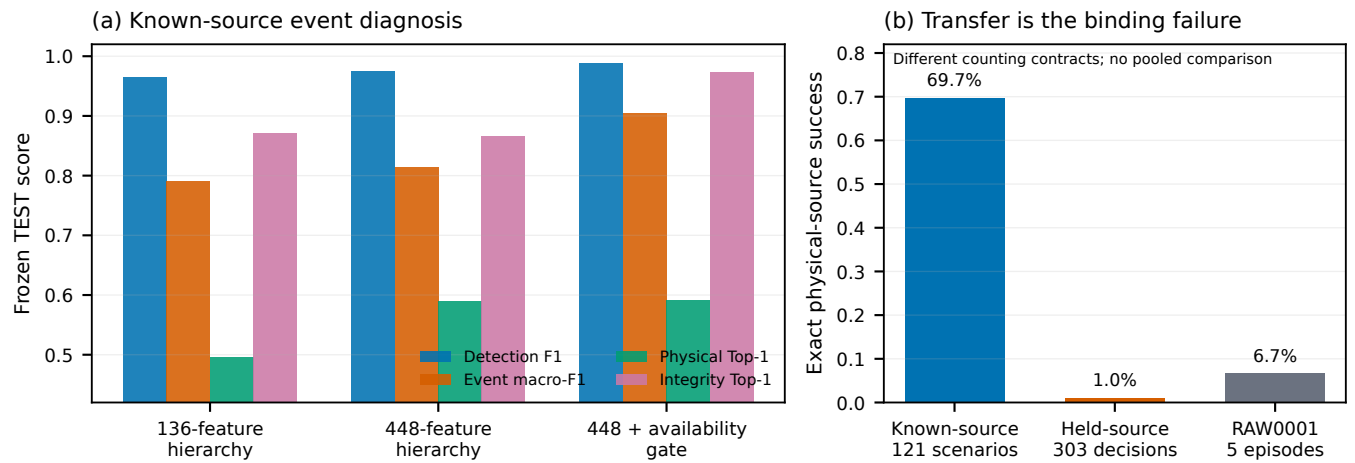


FIGURE 3. Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.

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